

Application Form Vrije Universiteit Amsterdam

Application to a Master's Degree programme for International Degree Holders

Instructions: Please put all the required documents in this format. Read the instructions first, then delete the text and insert the requested documents. When you have added all documents, upload the form in **.PDF format** in the *Complete Registration* section of [your personal dashboard](#) on vu.nl. Only submit an application file that includes all required documents in this template.

Details regarding your *previous or current* education

If you have completed multiple degrees at university level (Bachelor and Master), make sure to mention all of them and number them accordingly.

Country of your previous education	China
Name of the university	Beijing Sport University
Title of the degree(s) in English as well as in the original language (for example: Bachelor, License, Laurea, Mestre, etc.)	Bachelor, 本科学士学位
Subject of your degree (for example Bachelor in Computer Science)	Bachelor in Data Science and Big Data Technology
The (expected) date of graduation	June 2023
Nominal duration of your programme(s) / degree(s) in years (number of years to complete your programme)	4 years
Total credits needed to complete your degree (no need to convert it to ECTS)	172 credits
Is a thesis part of your curriculum?	Yes
Your current or final GPA (Grade Point Average), also include the maximum possible GPA in your university's system (for example: 8/10, 3.27/4, etc.)	88.1/100
English language test results (if not obtained yet, please indicate the expected date of submission)	IELTS Academic Overall: 6.5 (Listening: 7.0, Reading:6.5, Writing:6.5, Speaking:6.5)

Curriculum Vitae (CV)

A curriculum vitae that includes your personal details and lists your previous education and relevant work experience. You can for example use [an Europass CV template](#).

ZHOU Weiyi

(+86) 13270175603 | zhouwy2023@163.com

EDUCATION

Beijing Sport University, Beijing, China Sep. 2019- Jun. 2023

B.E. in Data Science and Big Data Technology, GPA: 88.1/100

Core modules: Mathematics; Statistics; Computer Theory; Data Structure; Algorithms; Programming; Database; Information Processing; Machine Learning, etc.

ACADEMIC PRACTICAL EXPERIENCES

Undergraduate Thesis Project, Beijing Sport University

Project: "Recognition and Evaluation of Yoga Poses Based on Deep Learning" Jun. 2022-Jun. 2023

Project Leader of Individual Project

- Conducted extensive research and literature review in the field, synthesizing relevant scholarly materials to provide a comprehensive literature review;
- Accomplished the collection and annotation of a diverse dataset comprising 1,073 images, meticulously categorized into 10 distinct classes;
- Designed and implemented three fusion strategies utilizing Convolutional Neural Networks (VGG) and Multilayer Perceptrons (MLP) to process both RGB image data and human keypoint coordinate data extracted using MediaPipe. These multi-modal models were designed for yoga pose recognition and assessment;
- Executed rigorous experiments and validation tests to assess the effectiveness of the multi-modal models; Identified the top-performing model based on accuracy as the ultimate solution;
- Achieved an outstanding recognition model accuracy rate of 99%; Attained accuracy rates ranging from 69% to 80% for five distinct evaluation models;
- Completed the entire thesis writing.

National Mathematical Modeling Competition

Project: "Procurement and Transportation Plan for Company Raw Materials" Sep. 2021

Core Member

- Developed a core indicator system for assessing supplier delivery reliability and created three mathematical models collaborating with the team. Utilized methods such as Analytic Hierarchy Process (AHP), Integer Programming, and Multi-Objective Programming;
- Contributed significantly in proposing the modeling approaches, took the lead in constructing one of the models, and completed the majority of the thesis writing;
- Awarded the Second Prize in the Beijing Municipal Competition.

National Computer Design Competition

Project: "Data Visualization for Paralympic Sports" Apr. 2021-Aug. 2021

Core Member

- Designed visually appealing and interactive data visualizations with a focus on practicality, clarity, completeness, and aesthetics to present information effectively collaborating with the team;
- Played a key role in data cleaning and integration (using Excel/Python) and utilized Python-based pycharts and JavaScript-based Echarts for data visualization; Contributed to the creation of a data visualization website;
- Awarded the Third Prize at the national level.

WORK EXPERIENCES

Beijing TAL Books Co., Ltd., Beijing, China

Mar. 2023- Jun. 2023

Intern, Data Engineering

- Generated weekly and monthly profit and loss reports (TikTok Channel Sales), along with influencer performance reports, through data extracted from the company's MySQL and PostgreSQL databases. Segmented data (using Tableau and SQL) based on diverse dimensions, effectively meeting evolving business demands and real-world scenarios;
- Responsively handled real-time data requests from business analysts by creating and maintaining Tableau real-time data dashboards and producing data files;
- Gathered industry-related data, executed data cleaning, processing, and visualization tasks, and conducted comprehensive brand comparisons across different categories at different time intervals, to provide valuable insights into market trends and competitive intelligence.

Beijing Kingsoft Office Software, Inc., Beijing, China

Sep. 2022- Dec. 2022

Intern, Data Analysis

- Conducted data cleansing, querying, and other processing tasks using Python's pandas (along with os and numpy modules) and Excel functions. Generated over 100 data files, including large files containing hundreds of thousands of records;
- Automated three routine data processing tasks through Python programming, including multi-channel enterprise information join queries and corrections, monthly client information update synchronization, and enterprise-type classification. These automation initiatives reduced manual processing time by approximately 2 hours per task, significantly enhancing departmental productivity;
- Facilitated daily communication and collaboration with sales and HR departments. This involved understanding their data-related requirements, such as data queries and database updates related to customers, orders, and employee website accounts, and providing prompt responses upon task completion;
- Performed routine updates to CRM data systems based on specific requirements, and tested the effectiveness of upcoming CRM system features.

SELECTED AWARDS AND HONORS

The Second Prize Scholarship, Beijing Sport University

Dec. 2022

The Second Prize Scholarship, Beijing Sport University

Dec. 2021

The Third Prize Scholarship, Beijing Sport University

Dec. 2020

Merit Student, Beijing Sport University

Sep. 2022

Merit Student, Beijing Sport University

Sep. 2020

LANGUAGES & SKILLS

Languages: Mandarin (Native), English (IELTS: 6.5)

Programming Languages:

- Proficient: SQL, Python, MATLAB, Spark;
- Command of syntax, logic, and architecture in: C, C++, JavaScript, and some mainstream NoSQL database languages;
- Experienced with Linux systems and distributed platforms (Hadoop, Hive, HBase).

Programming Applications: (Proficient in) Data Analysis, Data Visualization, Computer Vision, Deep Learning; (Familiar with) Digital Image Processing, Front-end Development.

Other Computer Skills: MS Office Suite (Proficient in Excel Functions and Pivot Tables), SPSS, Tableau.

Statement of Intent (Letter of Motivation)

A 1-2 page letter stating your reasons for applying to the programme (if applicable, please specify the **specialization** you are applying for).

Dear Admissions Committee,

I am writing to express my enthusiastic interest in the joint master's program in Computer Science offered by the University of Amsterdam in collaboration with Vrije Universiteit Amsterdam. My passion lies in advancing my academic and professional journey in the field of big data engineering, and I firmly believe that this program presents an exceptional opportunity for me to achieve this objective.

During my undergraduate studies at Beijing Sport University, I pursued a bachelor's degree in Data Science and Big Data Technology. My academic journey was enriched by a diverse range of courses, including Probability Theory and Mathematical Statistics, Data Structure and Algorithm Analysis, Python Programming, Principles and Applications of Databases, Machine Learning, and Neural Network Design and Programming. These courses spanned various domains, including Mathematics, Computer Science Foundations, and Data and Information Systems. Through these courses, I developed a profound appreciation for the transformative power of data in decision-making, problem-solving, and its contributions to various industries.

My specialization in data science and big data technology equipped me with a robust foundation in statistics, programming, databases, and machine learning. To apply this theoretical knowledge in real-world scenarios, I pursued internships. During my internship as a data analyst at KingSoft Office, I discovered my passion for implementing data processing tasks through programming, particularly in larger-scale projects. These tasks often received recognition from my superiors, and I frequently provided valuable guidance to fellow interns facing programming challenges. This internship experience solidified my desire to pursue a career in data-related programming and filled me with enthusiasm and confidence. As a result, I am eager to further my studies in the field of data engineering.

The Big Data Engineering track of this program places a strong emphasis on data processing infrastructure and offers technology solutions spanning various enterprise scales and application scenarios. It focuses on researching technologies to effectively handle large-scale data challenges. These attributes align perfectly with the current demands in the field of data management. Consequently, this program provides abundant educational resources and essential technical knowledge for pursuing a career in data engineering.

The joint Master's program in Computer Science boasts a distinguished faculty with renowned researchers from both VU and UvA. This collaboration creates one of Europe's most robust and

distinctive Computer Science programs, offering students access to outstanding educational resources from two prestigious universities simultaneously. The curriculum covers a wide spectrum of cutting-edge topics, emphasizing practical application and a commitment to addressing real-world challenges.

The curriculum of the Big Data Engineering track encompasses a broad range of subjects, from fundamental principles in algorithms and system theory to practical application technologies like data extraction, mining, analysis, and protection. Additionally, it includes coursework designed to cultivate research skills, helping students solidify their academic capabilities. The four compulsory courses in the first year—Data Protection Technologies, Web Data Processing Systems, Data Preparation, and Data Mining Techniques—provide a comprehensive skill set, covering critical aspects such as ETL (Extract, Transform, Load), Data Security and Compliance, and Data Modeling skills essential for aspiring data engineers. Furthermore, the program offers a variety of elective courses spanning diverse domains, including Foundations of Computing and Concurrency, Mathematics, Software Engineering, Societal Perspectives on Computer Science, Security, and Research Skills. This flexibility empowers students to tailor their studies to align with their specific goals and requirements, significantly enhancing their specialization in Big Data Engineering.

I believe that these course offerings will equip me with a versatile skill set and comprehensive knowledge that will prove invaluable in my career as a data engineer. During my internship at TAL Books, I had the privilege of working closely with a skilled Data Engineer named Wang. Wang's primary responsibilities revolved around ETL tasks. His extensive knowledge, strong work ethic, and unwavering commitment to continuous learning allowed him to consistently meet the team's requirements with speed and exceptional quality. His ability to efficiently address the team's demands garnered admiration and respect from all of us. Engaging in conversations with Wang and gaining insight into his job responsibilities and skills highlighted the importance of deepening my knowledge in key areas such as Web Data Extraction and Advanced Algorithms. Recognizing this need, I am committed to further enhancing my expertise in these domains to prepare myself for the future responsibilities of an independent data engineer.

In conclusion, I am convinced that the joint master's program in Computer Science is the perfect platform for me to further specialize in computer science, conduct cutting-edge research, and prepare for a fulfilling career in data engineering. I am committed to bringing my passion, specialized skills that align with the program's requirements, and enthusiasm for computer science to this program. I am excited about the possibility of joining the esteemed University of Amsterdam and Vrije Universiteit Amsterdam community.

Thank you for considering my application. I eagerly anticipate the opportunity to contribute to and learn from the exceptional faculty and fellow students in this program. If granted admission, I am confident that I will excel academically and make a positive contribution to the field of computer science.

Sincerely,

Zhou Weiyi

English language test results

***Optional** document at this stage of the application.*

You can already apply without having the English language test results and you do **not** need to include them in this application template.

If you already have **valid and sufficient** test results, please **upload** them via your VU personal dashboard in the separate category.

The **final deadline** for sending us the English Language test is **1 June (non-EEA) and 31 August (EEA)**. However, we **recommend** students to not wait until the final deadline and to make sure to hand in their English Language test no later than **1 May (non-EEA) and 1 August (EEA)**.

Please note that the minimum required scores can **differ** per Master's programme. For all relevant information on our English language test requirements, please visit [this page](#).

Diploma

Optional document at this stage of the application.

If you have already graduated, please insert a copy of your original Bachelor's diploma. Please include all relevant diplomas if you have multiple Bachelor's and/or Master's diplomas.

If you are conditionally admitted, the final deadline for sending us the **certified copy** of your diploma via postal mail is 31 August. We recommend students to hand in this document well in advance. Full details on handing in the certified copy of your diploma can be found on [this page](#).



Transcript of records

Please insert your (most up to date) **official transcript of records** from your Bachelor's and/or Master's programme here. If your university cannot provide a transcript in English, then please add your own translation as well.

In case you went on **exchange** during your programme and the courses and grades are not mentioned on your main transcript, please add the transcript from your exchange university as well.

Please include a grading system explanation.

Beijing Sport University
Student's Academic Record

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Name	Zhou Weiyi		Student number	2019012628		School	School of Sports Engineering			
Gender	Female					Specialty	Data Science and Big Data Technology			
Date of Birth	20001110		Degree			Degree of Bachelor of Engineering	School Year	4 years		
Date of Enrollment	20190901		Date of Graduation	20230618						
Course	Exam Date	Type	Credit	Score	Course	Exam Date	Type	Credit	Score	
Career Planning for College Students	2019-2020-1	R	1	94	Python Programming	2020-2021-2	E	2	83	
College English B1	2019-2020-1	R	4	98	Operating System	2020-2021-2	R	3	79	
Advanced Mathematics I	2019-2020-1	R	6	89	Multivariate Statistical Analysis	2020-2021-2	R	3	82	
Introduction to Computer	2019-2020-1	R	2	97	Computer Network	2020-2021-2	R	3	86	
Military Training	2019-2020-1	Good	2		Visual Programming	2020-2021-2	E	1	97	
Sports-Sanda	2019-2020-1	R	2.5	96	Introduction to Basic Principles of Marxism	2020-2021-2	R	3	84	
Linear Algebra	2019-2020-1	R	3	78	Introduction to Bridge	2020-2021-2	E	2	93	
Writing and Mind Training	2019-2020-1	R	2	81	Principle and Application of Database	2020-2021-2	R	3	92	
Introduction to Intelligent Sports	2019-2020-1	R	2	88	Digital Signal Processing	2020-2021-2	R	3	82	
Outline of Modern and Contemporary Chinese History	2019-2020-1	R	3	70	Virtual Simulation Experiment Course-2	2020-2021-2	E	1	Pass	
Human Anatomy and Physiology	2019-2020-2	R	4	93	Operational Research	2020-2021-2	E	3	90	
Programming C Language	2019-2020-2	R	3	90	Sports Biomechanics	2020-2021-2	E	3	85	
Introduction to College Students Entrepreneurship and Practice (Online)	2019-2020-2	E	1	88	NoSQL Database Technology	2021-2022-1	E	3	81	
Education on Psychological Health for College Students	2019-2020-2	R	1	89	Aesthetic Labor (Online)	2021-2022-1	E	1	98	
College English B2	2019-2020-2	R	4	94	Principle and Application of Big Data Technology	2021-2022-1	R	3	92	
Foundation of Circuit and Electronic Technology	2019-2020-2	R	3	78	CET-6	2021-2022-1	Others	0	498	
Advanced Mathematics II	2019-2020-2	R	6	84	Machine Learning	2021-2022-1	R	3	89	
19th Century European Writers and Works (Online)	2019-2020-2	E	1	92	Labor Education (Online)	2021-2022-1	E	1	99	
Ideological Morality Cultivation and Law Foundation	2019-2020-2	R	3	88	Digital Image Processing	2021-2022-1	R	3	88	
Sports-Tai Chi, Sword	2019-2020-2	R	2.5	95	Method and Application of Sports Data Statistics	2021-2022-1	R	3	96	
Office Advanced Application	2020-2021-1	E	2	93	Sports Performance Analysis	2021-2022-1	R	3	90	
Infectious Disease and Public Health	2020-2021-1	E	1	86	Introduction to Principles of Exercise and Fitness	2021-2022-1	E	1	86	
College English B3	2020-2021-1	R	4	92	Spark Programming Foundation	2021-2022-2	E	3	92	
CET-4	2020-2021-1	Others	0	590	College Students Employment Guidance	2021-2022-2	R	1	92	
Probability Theory and Mathematical Statistics	2020-2021-1	R	4	68	Aesthetic Education-Body Shape Training	2021-2022-2	E	1.5	97	
Principles of Computer Composition	2020-2021-1	R	3	84	Aesthetic Education-Structure Beauty-Spatial Shape Design (Online)	2021-2022-2	E	1	100	
Military Theory	2020-2021-1	R	2	97	Interpretation of Civil Code	2021-2022-2	E	1	89	
Discrete Mathematics	2020-2021-1	R	3	92	Neural Network Design and Programming	2021-2022-2	E	2	91	
Introduction to Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics	2020-2021-1	R	5	87	Web Crawler and Data Collection	2021-2022-2	E	2	91	
Object-oriented Technology	2020-2021-1	R	3	86	Graduation Thesis	2022-2023-2	Graduation	4	92	
Data Structure and Algorithm Analysis	2020-2021-1	R	3	86	Graduation Practice	2022-2023-2	Graduation	6	97	
Introduction to Sports B	2020-2021-1	R	2	93	Innovation and Entrepreneurship Project	2022-2023-2	Graduation	4	89	
New Media and Social Sex (Online)	2020-2021-1	E	1	99	Social Investigation	2022-2023-2	Graduation	3	90	
Motion Learning and Control	2020-2021-1	R	3	95	Situation and Policy	2022-2023-2	Graduation	2	90	
Total credits in All Academic Years	173.5		Overall credit	172		GPA	3.81		Official Seal	

NOTE: R: Required Course E: Elective Course

Lister:  

Date: 2023/6/19 



北京体育大学教务处

Teaching Affairs Office of Beijing Sport University

课程考核和成绩记载

学校采用绩点制作为评价学生各类课程和部分教学实践环节（毕业论文（设计）和专业实习）学习质量的成绩评价制度。

（一）等级、绩点、百分制成绩对应关系表如下：

字母等级	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	F
中文等级	优秀			良好			中等			合格		不合格
绩点	5.0~4.7	4.6~4.4	4.3~4.0	3.9~3.7	3.6~3.4	3.3~3.0	2.9~2.7	2.6~2.4	2.3~2.0	1.9~1.1	1	0
百分制	100~97	96~94	93~90	89~87	86~84	83~80	79~77	76~74	73~70	69~61	60	<60

（二）学分绩点数

学分绩点数按学期、学年、四学年等分别结算，方法如下：

学分绩点数 = $\sum$ （取得每门课程的学分数 × 取得该门课程的绩点数）

平均学分绩点数 = 学分绩点数 / 学生所修课程的学分数（计到小数点后二位）

（三）通识必修课、专业必修课、专业选修课、实践环节中的毕业论文（设计）和专业实习参与绩点计算。

Grade Calculation Rules

The school uses GPA as the evaluation system to estimate the learning quality and quantity of a student's achievement throughout various courses as well as some practical learning experiences (thesis, education practice, internships etc.).

(A) The corresponding relation between one's grades, grade point and the percentage system is as follows:

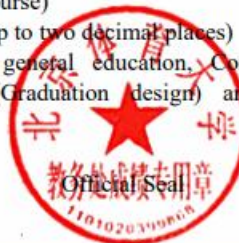
Letter grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Chinese level	excellent			good			average			pass		fail
GPA	5.0~4.7	4.6~4.4	4.3~4.0	3.9~3.7	3.6~3.4	3.3~3.0	2.9~2.7	2.6~2.4	2.3~2.0	1.9~1.1	1	0
Centesimal system	100~97	96~94	93~90	89~87	86~84	83~80	79~77	76~74	73~70	69~61	60	<60

(B) GPA is calculated as follows:

Overall GPA = $\sum$ (credit of each course × GPA of that course)

Average GPA = Overall GPA / Overall credit (counting up to two decimal places)

(C) GPA calculation includes: Compulsory courses of general education, Core required courses, Subject elective courses, Thesis (Graduation design) and Internships.



Course description including used literature

Details regarding your previous or current education	
<i>Please do not submit your programme's official syllabus/curriculum overview.</i>	
Name programme(s)/degree(s)	Bachelor of Engineering in Data Science and Big Data Technology
Nominal duration of your programme(s) / degree(s) in years	4 years
Total credits needed to complete your degree (no need to convert it to ECTS)	172 credits

Please describe the courses you have taken during your previous higher education that are relevant to the Master's programme you are applying for.

The **relevant courses for admission** can be found on the [programme pages](#), under the **tab 'Admissions'**, section **'I have an international degree'**.

Also mention the relevant courses you will still complete before graduation.

Relevant courses for intended Master's programme		
Course name	Advanced Mathematics I & II	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 1st Year	
Credits or study hours	<u> 6+6=12 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course covers a wide array of mathematical concepts, starting with the fundamentals of functions with multiple variables. Students will explore key topics like limits, continuity, partial derivatives, and complete differentials, along with practical applications using the chain rule. In addition, the curriculum delves into more advanced subjects, including Taylor's formula for multivariable functions, the theorem of implicit function existence, and the intricate nuances of extreme value problems. Its primary objectives	

	are twofold: first, to provide students with a comprehensive grasp of calculus and differential equations; second, to foster critical thinking, abstract reasoning, and advanced arithmetic skills. Throughout the course, students will engage with a diverse range of mathematical ideas, building a strong foundation for their academic pursuits.	
Used literature / textbook	1. Department of Mathematics of Tongji University. (2014). Advanced Mathematics, 7th Edition (Volume 1). Higher Education Press. 2. Department of Mathematics of Tongji University. (2014). Advanced Mathematics, 7th Edition (Volume 2). Higher Education Press.	
Course name	Linear Algebra	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 1st Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course comprehensively explores essential linear algebra topics, including matrix theory, determinants, linear simultaneous equations, linear spaces, transformations, similar matrices, and quadratic forms. Its primary objectives revolve around providing students with a robust theoretical foundation in linear algebra, fostering their abilities in abstract thinking and logical reasoning, and honing their proficiency in matrix operations and computational techniques. Furthermore, the course aims to empower students to apply their knowledge of linear algebra to solve real-world problems, utilizing computer-based tools for support. Upon completing this course, students will not only possess a comprehensive understanding of the core principles of linear algebra but will also have developed their skills in abstract thinking and logical reasoning. These acquired capabilities will prepare them for future academic and practical pursuits, enabling them to approach complex challenges with confidence and competence.	
Used literature / textbook	Department of Mathematics of Tongji University. (2014). Linear Algebra, 7th Edition. Higher Education Press.	
Course name	Probability Theory and Mathematical Statistics	

Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 4 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course encompasses a wide range of topics, including random experiments, events within sample spaces, stochastic variables, bivariate random variables and their distributions, as well as fundamental concepts in mathematical statistics. Its core objectives are to provide students with a comprehensive and in-depth understanding of probability theory and mathematical statistics, ensuring they acquire essential knowledge, theories, and methodologies in these fields. Additionally, the course is designed to nurture students' analytical thinking and problem-solving skills, empowering them to address real-world challenges using probabilistic statistical approaches. By the end of this course, students will have developed a solid foundation in probability theory and mathematical statistics, equipping them with the skills and insights necessary to comprehend and apply these principles effectively. This comprehensive understanding will serve as a valuable asset as they pursue further studies or apply their expertise to practical scenarios, enabling them to make informed decisions and contributions in various domains.	
Used literature / textbook	Mao Shisong, Cheng Yiming, Pu Xiaolong. (2019). Course on Probability Theory and Mathematical Statistics. Higher Education Press.	
Course name	Discrete Mathematics	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or class hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	Discrete Mathematics is a foundational course specifically tailored for majors in computer science and technology. This course has a twofold objective: to equip students with a comprehensive understanding of the concepts and theories within discrete mathematics and to enable them to apply these tools effectively to analyze and solve real-world problems. It emphasizes the development of abstract thinking and logical reasoning skills, essential for success in computer science. The contents of this course	

	are mathematical logic, counting, set theory, relational theories, modern algebra, graphics and combinatorial mathematics. Upon completing this course, students will not only possess a strong theoretical foundation in discrete mathematics but will also have honed their abstract thinking and logical reasoning abilities.	
Used literature / textbook	Geng Suyun, Qu Wanling, Zhang Liang. (2013). Discrete Mathematics (Fifth Edition). Tsinghua University Press.	
Course name	Multivariate Statistical Analysis	
Bachelor year (1, 2, 3 etc.), Master year or Course level (Introductory, Intermediate or Advanced)	Bachelor, 2nd Year	
Credits or study hours	___ 3 ___ credits	_____ study hours
Course description or learning outcomes (5-10 sentences)	This course delves into the realm of multivariate statistical analysis, encompassing topics such as multivariate normal distribution, data visualization techniques, cluster analysis, discriminant analysis, principal components analysis, factor analyses, correspondence analysis, canonical analysis, multivariate regression analyses, and the statistical examination of qualitative information. With a dual focus, the course aims to achieve two key objectives: first, to furnish students with a comprehensive grasp of diverse statistical methods, including their foundational principles and practical applications; and second, to bolster their communication capabilities, encompassing both oral presentations and written reports. Through this curriculum, students will cultivate the ability to address complex real-world problems employing multivariate statistical tools. The synergy of technical proficiency and communication skills equips them to dissect intricate datasets and convey their findings with precision and clarity.	
Used literature / textbook	Richard A. Johnson, Dean W. Wichern. (2012). Applied Multivariate Statistical Analysis (6/e). Phi Learning Private Limited.	
Course name	Operational Research	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	___ 3 ___ credits	_____ study hours

Course description or learning outcomes (5-10 sentences)	The course covers basic philosophy and concepts of operational research, linear programming, transportation problems, queuing theory, decision analysis, simplex methods, network analysis, random processes and dynamic programming. The primary objectives of this course are to provide students with a comprehensive understanding of operational research's development, principles, and quantitative problem-solving techniques. Through this course, students will acquire the skills necessary to address complex managerial issues involving multiple variables. Additionally, they will enhance their decision-making abilities by harnessing quantitative methods and leveraging computer skills. Upon successful completion of this course, students will possess a comprehensive knowledge of operational research, equipping them with the skills and methodologies required to tackle intricate managerial problems involving multiple variables. They will also have the ability to make informed quantitative decisions, supported by computer skills and specialized knowledge.	
Used literature / textbook	Hu Yunquan, Guo Yaohuang. (2011). Operations Research (5th Edition). Tsinghua University Press.	
Course name	Method and Application of Sports Data Statistics	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year	
Credits or study hours	3 credits	study hours
Course description or learning outcomes (5-10 sentences)	This course covers the fundamental concepts and principles of sports statistics, including data collection, examination, and organization. It also delves into statistical measures, ratios, probability, and common probability distributions. Students will gain hands-on experience with SPSS software for data analysis and presentation. Upon completion of the course, students will be able to understand the core principles of sports statistics, apply statistical techniques to sports data, and effectively utilize SPSS software for data analysis and presentation. They will also develop the skills to interpret ratios, probabilities, and probability distributions in the context of sports statistics, enabling them	

	to make data-driven decisions and evaluations.	
Used literature / textbook	Quan Deqing. (2011). Sports University Common Textbook: Sports Statistics. Renmin Sports Publishing House.	
Course name	Programming C Language	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 1st Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course introduces the fundamentals of computer programming and the C programming language. It covers topics such as the basics of computer programs, the evolution and characteristics of the C language, the structure of C programs, and the steps to run C programs. Additionally, it explores algorithms, data structures, data types, operators, control structures, and arrays. Upon completing this course, students will be able to understand the essentials of computer programming and the C language, including program structure and execution. They will learn to design and implement basic algorithms, use data structures effectively, and develop modular and structured programs. Students will also gain hands-on experience in writing and running C programs, as well as problem-solving skills through programming exercises and assignments.	
Used literature / textbook	Tan Haoqiang. (2017). C Programming (Fifth Edition). Tsinghua University Press.	
Course name	Object-oriented Technology	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course covers fundamental C++ concepts, including program structure, data types, control flow, functions, and classes. It delves into advanced topics like operator overloading, inheritance, and polymorphism, providing a comprehensive understanding of C++ programming. Practice in C++ programming are also included in the course. By the end of this course, students will have acquired strong C++ programming skills, enabling them to design and implement complex software solutions. They will gain proficiency in object-oriented programming and learn to apply advanced C++ features,	

	enhancing their ability to develop efficient and maintainable software.	
Used literature / textbook	Tan Haoqiang. (2014). C++ Object-Oriented Programming (2nd Edition). Tsinghua University Press.	
Course name	Python Programming	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 2 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	In the first part of this course, students will be introduced to the Python programming language, covering fundamental concepts such as program design, basic syntax, and the development environment. The second part delves deeper into Python, exploring topics like data types, control structures, functions, and code reuse. Finally, the third part focuses on practical applications, including scientific computing, data visualization, web scraping, and automation. In the first part of this course, students will be introduced to the Python programming language, covering fundamental concepts such as program design, basic syntax, and the development environment. The second part delves deeper into Python, exploring topics like data types, control structures, functions, and code reuse. Finally, the third part focuses on practical applications, including scientific computing, data visualization, web scraping, and automation.	
Used literature / textbook	Song Tian, Li Xin, Huang Tianyu. (2017) . Python Language Program Design Fundamentals (2nd Edition). Higher Education Press.	
Course name	Data Structure and Algorithm Analysis	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course covers fundamental concepts in data structures and algorithm analysis. It starts with an introduction to data structures, abstract data types, algorithms, and algorithm analysis. The course then explores linear lists, including sequential and linked representations, as well as stack and queue data structures with practical applications. Additionally, it delves into strings, arrays, and generalized lists, followed by an introduction to trees, binary trees, and tree traversal techniques. By the end of this course,	

	students will have a solid understanding of data structures and their representations, including linear lists, stacks, queues, strings, arrays, and generalized lists. They will also be able to apply various data structures in practical scenarios and develop efficient algorithms. Additionally, students will gain proficiency in tree structures and their traversal, along with the ability to solve problems related to trees and graphs. This course will equip students with essential skills for algorithm analysis and problem-solving in computer science and related fields.	
Used literature / textbook	Yan Weimin, Wu Weimin. (1997). Data Structures in C Language. Tsinghua University Press.	
Course name	Machine Learning	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course on Machine Learning covers a wide range of topics in the field, starting with an introduction to machine learning concepts and the historical development of the field. It progresses to cover essential concepts such as model evaluation, linear models, decision trees, neural networks, support vector machines, Bayesian classifiers, and ensemble learning. Each chapter dives deep into these topics, providing a comprehensive understanding of machine learning algorithms and techniques. By the end of this course, students will be able to grasp fundamental machine learning concepts, evaluate and select appropriate machine learning models for various tasks, and understand the inner workings of algorithms like linear regression, decision trees, neural networks, support vector machines, and Bayesian classifiers. Additionally, students will learn how to apply these machine learning techniques to real-world problems and gain valuable experience in model evaluation and selection.	
Used literature / textbook	Zhou Zhihua. (2016). Machine Learning. Tsinghua University Press.	
Course name	Introduction to Computer	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 1st Year	

Credits or study hours	__2__ credits	_____study hours
Course description or learning outcomes (5-10 sentences)	The "Introduction to Computer" course covers a comprehensive spectrum of computer science topics, equipping students with essential knowledge and skills. It commences by teaching problem-solving through computer applications, emphasizing modeling and abstraction. Students delve into computer information digitization, encompassing number systems, binary representation, character encoding, and multimedia data handling. The course then delves into computer hardware architecture, tracing the evolution of computing devices and explaining their core components. Additionally, students explore software platforms, including operating systems, data storage, and program execution management. It provides insights into computer networks, the Internet, information security, cloud computing, and the Internet of Things (IoT). The curriculum covers data processing, databases, algorithms, data structures, and fundamental programming principles. By the course's conclusion, students will have acquired a strong foundation in computer science, enabling them to understand, navigate, and leverage the ever-evolving world of computing.	
Used literature / textbook	Li Fengxia, Chen Yufeng, Shi Shumin. (2016). College Computer. Higher Education Press.	
Course name	Foundation of Circuit and Electronic Technology	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 1st Year	
Credits or study hours	__3__ credits	_____study hours
Course description or learning outcomes (5-10 sentences)	This course, "Foundation of Circuit and Electronic Technology," explores the essential principles of circuit analysis, analog electronics, and digital logic circuits. Covering a wide array of topics, it delves into circuit fundamentals, including key concepts, basic laws, and circuit elements. Students will learn techniques to analyze resistive circuits, understand circuit theorems, and gain proficiency in analyzing dynamic circuits. The course also provides insights into AC circuit analysis with a focus on sinusoidal signals. Furthermore, it introduces semiconductor devices, operational amplifiers, and digital logic gates. By the end of this course, students will	

	possess a strong foundation in electrical engineering and electronics, enabling them to excel in advanced coursework and practical applications in these fields.	
Used literature / textbook	Li Xinguang, Wang Jinkuang, Zhang Jing. (2016). Fundamentals of Circuits and Electronics, 2nd Edition. China Machine Press.	
Course name	Principles of Computer Composition	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	In the "Introduction to Computer" course, students will embark on a comprehensive journey through the fundamental principles of computer systems. Starting with an exploration of the historical development and real-world applications of computers, the curriculum delves into the core hardware components, including the central processing unit (CPU), memory, and input/output systems. It also covers essential topics like system buses, machine instructions, addressing modes, and instruction formats. Additionally, students will gain insights into control unit design and microprogramming techniques. The course further examines input/output systems, including I/O devices, interfaces, and data transfer methods, as well as program execution methods and interrupt mechanisms. Understanding direct memory access (DMA) and the broader societal impact of computers is also a key focus. By the end of this course, students will have a solid foundation in computer systems, setting the stage for further studies in computer science and related fields.	
Used literature / textbook	Tang Shuofei. (2008) . Principles of Computer Composition (2nd Edition). Higher Education Press.	
Course name	Operating System	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course provides a comprehensive overview of operating system fundamentals. Covering a wide range of topics, the course introduces students to the goals, development, characteristics, and primary functions of operating systems. It explores process	

	management, including process description, control, synchronization, and communication, as well as memory management, encompassing storage hierarchy, program loading, memory allocation, and virtual memory. The course also delves into input and output systems, discussing I/O devices, controllers, interrupts, and device drivers. File and disk storage management, system interfaces, and security measures are important components of the curriculum. Additionally, students are introduced to advanced subjects like multi-processor systems, multimedia operating systems, and trusted systems. By course completion, students will have a strong foundational understanding of computer operating systems, equipping them to comprehend the pivotal role these systems play in contemporary computing and preparing them for further advanced studies in this field.	
Used literature / textbook	Tang Xiaodan, Liang Hongbing, Zhe Fengping, Tang Ziyang. (2014). Computer Operating Systems (4th Edition). Xi'an University of Electronic Science and Technology Press.	
Course name	Computer Network	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	The course "Computer Network" offers a comprehensive exploration of computer networks in the context of the contemporary information era. Starting with an overview of the role and development of computer networks, it progresses into the fundamental components and categories of these networks. It covers essential topics such as the physical layer, data link layer, and network layer, examining protocols like TCP/IP and IP addressing in depth. The course then delves into transport layer protocols like UDP and TCP, emphasizing their roles in reliable data transmission. It also explores application layer services, including DNS, FTP, HTTP, and email protocols. Additionally, the curriculum addresses advanced concepts such as network security, cryptography, firewalls, and intrusion detection systems. Throughout the course, students gain valuable insights into wireless and mobile networks, understanding the principles	

	behind WLANs, WPANs, WMANs, and cellular communication technologies like GSM and Mobile IP. By the course's conclusion, students will have acquired the knowledge and skills necessary for careers in network administration, cybersecurity, or network engineering, enabling them to contribute effectively to the ever-evolving world of computer networks.	
Used literature / textbook	Xie Xiren. (2017). Computer Networks (7th Edition). Electronic Industry Press.	
Course name	Principle and Application of Database	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course, "Principle and Application of Database," provides a comprehensive exploration of database systems. Starting with an overview of databases, data models, and the structure and components of database systems, students will gain a solid foundation. The course delves into the theory and practice of relational databases, including formal definitions, operations, integrity, and the SQL language. As students progress, they'll learn about important aspects of database security, integrity, and principles of data normalization. They will also delve into the intricacies of database design, including requirements analysis, conceptual and logical design, and physical implementation. Additionally, students will explore database programming techniques, including SQL, embedded SQL, stored procedures, and more. The course covers advanced topics such as query processing and optimization, database recovery techniques, and concurrency control. Students will also gain insights into the architecture and functions of database management systems. Furthermore, the course addresses emerging trends in the field, including big data management, in-memory databases, data warehousing, and online analytical processing. It also introduces students to the concepts of data mining and data warehousing, which are crucial in today's data-driven world. By the end of this course, students will have a comprehensive understanding of database principles, design,	

	programming, and management, as well as insights into cutting-edge developments in the field. This knowledge equips them to excel in a wide range of database-related roles and adapt to the evolving landscape of data management.	
Used literature / textbook	Wang Shan, Sa Shixuan. (2014). Introduction to Database Systems (5th Edition). Higher Education Press.	
Course name	Digital Signal Processing	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 2nd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course, "Digital Signal Processing," delves into the fundamentals and advanced concepts of signal processing. The contents are structured into ten chapters covering discrete-time signals and systems, z-transform, discrete Fourier transform, fast Fourier transform, digital filter design, multirate signal processing, finite-word-length effects, and more. Throughout the course, students will explore topics such as convolution, system stability, and the design of both infinite impulse response (IIR) and finite impulse response (FIR) digital filters. The course also touches on the critical aspects of finite-word-length effects, quantization, and overflow, essential in practical digital signal processing applications. By the end of this course, students will have a comprehensive understanding of digital signal processing principles and techniques, equipping them with the knowledge and skills needed to analyze, design, and implement digital signal processing systems. Whether students are interested in audio processing, image analysis, or telecommunications, this course provides a solid foundation in the field, enabling them to apply DSP concepts effectively in various real-world scenarios.	
Used literature / textbook	Cheng Peiqing. (2017). Digital Signal Processing Tutorial (Fifth Edition) MATLAB Edition. Tsinghua University Press.	
Course name	Principle and Application of Big Data Technology	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year	
Credits or study hours	<u> 3 </u> credits	<u> </u> study hours
Course description or learning outcomes	"Principle and Application of Big Data	

(5-10 sentences)	Technology" is a comprehensive course designed to provide students with a profound understanding of big data concepts and practical skills in this rapidly evolving field. The course begins by introducing the era of big data, highlighting its societal impact and the technological advancements behind it, including batch and stream processing, graph computing, and query analysis. The curriculum extensively covers the Hadoop ecosystem, encompassing key components such as HDFS, HBase, MapReduce, Hive, Pig, Mahout, ZooKeeper, Flume, Sqoop, and Ambari. Practical knowledge is emphasized through hands-on installation and usage of Hadoop for distributed data processing. Furthermore, students delve into the intricacies of big data storage and management, exploring topics like distributed file systems, distributed databases, NoSQL databases, and cloud databases. The MapReduce programming model is dissected in detail, emphasizing its workflow and the critical shuffle process. The course also equips students with insights into optimizing Hadoop, addressing its limitations, and examining enhancements like HDFS 2.0 and the YARN resource management framework. Key components within the Hadoop ecosystem such as Pig, Tez, and Kafka are explored. With a focus on real-world applications, students gain exposure to data warehousing using Hive, advanced big data technologies like Spark, stream processing, Flink, graph computing, and the significance of data visualization. By the end of the course, students are well-prepared to leverage big data technologies across diverse professional domains, making data-driven decisions and contributing to the evolving field of big data.
Used literature / textbook	Lin Ziyu. (2021). Principle and Application of Big Data Technology (3rd Edition). People's Posts and Telecommunications Press.
Course name	Digital Image Processing
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year
Credits or study hours	<u> 3 </u> credits <u> </u> study hours
Course description or learning outcomes (5-10 sentences)	This course on "Digital Image Processing" provides a comprehensive overview of the

	subject matter, covering a wide range of topics essential for understanding and working with digital images. The course begins with an introduction to the fundamentals of digital image processing, including its origins, applications, and basic steps. Students will learn about key concepts such as image perception, sampling, and quantization, as well as the mathematical tools used in image processing. The course delves into various image processing techniques, including grayscale transformations, spatial filtering, and frequency domain filtering. Students will gain practical knowledge in enhancing and restoring images, reducing noise, and performing various operations in both the spatial and frequency domains. Color image processing is another vital aspect covered in the course, where students will explore color models, pseudocolor image processing, and techniques for working with full-color images. They will also learn about advanced topics like wavelet transformations and other image transformations. Furthermore, the course addresses image compression and watermarking, providing insights into various compression techniques and the concept of embedding information in digital images. Image segmentation and feature extraction are crucial components of this course, offering students the ability to partition images into meaningful regions and extract important characteristics for analysis. Finally, the course covers image pattern recognition and classification, introducing concepts like prototype matching, statistical classifiers, neural networks, and deep learning with convolutional neural networks (CNNs). By the end of this course, students will have a comprehensive understanding of digital image processing techniques and the skills necessary to apply them in various real-world applications.
Used literature / textbook	Rafael C. Gonzalez, Richard E. Woods. (2020). Digital Image Processing, Fourth Edition. Electronic Industry Press.
Course name	Neural Network Design and Programming
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year
Credits or study hours	<u> 2 </u> credits <u> </u> study hours
Course description or learning outcomes	"Neural Network Design and Programming" is a

(5-10 sentences)	comprehensive course that provides students with a deep dive into the world of deep learning, with a strong focus on practical implementation using PyTorch. Throughout this course, students will cover a wide range of topics, including the fundamentals of machine learning, data manipulation, linear algebra, and calculus. Students will also learn about key components like data, models, objective functions, and optimization algorithms. The course guides students through the implementation of linear neural networks, introducing concepts such as linear regression, softmax regression, and image classification using both manual coding and PyTorch. They'll progress to more complex topics like multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs). Modern deep learning techniques are a significant focus, with in-depth explorations of advanced architectures like GRUs, LSTMs, transformers, and attention mechanisms. Students will also delve into natural language processing, covering word embeddings, sentiment analysis, and text inference using pre-trained models like BERT. By the end of this course, students will have gained a strong foundation in deep learning theory and practical programming skills. They'll be equipped to tackle a wide range of applications, from image classification to natural language processing, using neural networks and PyTorch as their primary tools.
Used literature / textbook	Aston Zhang, Mu Li. (2021). Hands-On Deep Learning (PyTorch Edition). People's Posts and Telecommunications Press.
Course name	NoSQL Database Technology
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year
Credits or study hours	<u> 3 </u> credits <u> </u> study hours
Course description or learning outcomes (5-10 sentences)	The "NoSQL Database Technology" course offers a comprehensive exploration of modern database systems, focusing on the principles and practical applications of NoSQL databases. Beginning with an overview of traditional relational databases and their limitations, the course introduces the key features of NoSQL databases, emphasizing their adaptability to various data structures and

	scalability for big data challenges. Students will delve into the intricacies of distributed data management, studying concepts such as data sharding, replication, and eventual consistency. The course covers different NoSQL data models, including key-value, document-oriented, column-family, and graph databases, providing hands-on experience with popular NoSQL systems like HBase, Cassandra, MongoDB, and Neo4j. With a strong foundation in NoSQL principles, students will learn to design and implement database solutions tailored to specific use cases, mastering essential skills in database administration, configuration, and querying. Moreover, the course delves into advanced topics such as data visualization, security, and scalability, preparing students for real-world challenges in the ever-evolving field of data management. Upon completion, students will have a deep understanding of NoSQL databases, enabling them to make informed decisions about database selection and design for diverse applications, from web development to big data analytics. This course equips learners with valuable skills for careers in data engineering, database administration, and data-driven decision-making.
Used literature / textbook	Hou Bin. (2020). Principles of NoSQL database. People's Posts and Telecommunications Press.
Course name	Spark Programming Foundation
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year
Credits or study hours	<u> 3 </u> credits <u> </u> study hours
Course description or learning outcomes (5-10 sentences)	"Spark Programming Foundation" is a comprehensive course covering key concepts and practical applications of Spark, a prominent big data processing framework. This eight-chapter course begins with an introduction to big data technologies, Spark's design principles, and setting up and using Spark. It then delves into RDD programming, Spark SQL for data analysis, real-time data processing with Spark Streaming, and advanced structured streaming techniques. The course concludes with an exploration of machine learning using Spark MLlib. By completing this course, students will gain a thorough understanding of big data technologies,

	master Spark's fundamentals, and develop the skills to write and manage Spark applications. They will become proficient in working with RDDs, leverage Spark SQL for data analysis, and process real-time data with Spark Streaming and Structured Streaming. Additionally, students will learn to apply machine learning techniques using Spark MLlib, making them well-prepared for data engineering and data science roles in modern organizations.	
Used literature / textbook	Lin Ziyu. (2020). Fundamentals of Spark Programming (Python Edition). People's Posts and Telecommunications Press.	
Course name	Web Crawler and Data Collection	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	Bachelor, 3rd Year	
Credits or study hours	<u> 2 </u> credits	<u> </u> study hours
Course description or learning outcomes (5-10 sentences)	The course "Web Crawler and Data Collection" comprises four main parts. It initiates with an overview of big data collection's significance and Python's role in web scraping. The foundational section covers web page structure, encoding systems, and HTTP protocols. The technical implementation segment dives into basic and dynamic web scraping, web information extraction, and specialized topic-based crawling. Advanced concepts like Deep Web crawling, microblog data collection (using the Weibo platform), and anti-scraping techniques are explored. The course also addresses text information processing, vector space models, text classification, topic modeling, and data visualization for data mining. By the end, students will have the skills to apply web scraping in real-world scenarios such as news content analysis and SQL injection detection using web crawlers, making them adept in this essential field of data collection and analysis.	
Used literature / textbook	Zeng Jianping. (2020). Web Scraping and Big Data Collection and Mining. Tsinghua University Press.	

Thesis (or another sample of academic writing)

Please insert your **thesis** here or, if you did not write a thesis (yet), an **alternative writing sample** such as an academic paper or essay (written in English). If you have not written anything in English during your studies, then please select one of your own academic writing samples and add a 2-5 page summary in English **before** the text in the original language. Always include your **literature references**. Please be aware that for some programmes you need to upload a specific essay as mentioned on that programme's webpage.

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Summary

The thesis, titled "Recognition and Evaluation of Yoga Poses Based on Deep Learning", contributes to the field of yoga pose analysis. It accomplishes this by establishing the YogaCE dataset and developing neural network models, YogaRegNet and YogaEvaNet. In a world increasingly embracing the holistic benefits of yoga, the accurate recognition and assessment of yoga poses have become essential for both practitioners and instructors. This research endeavors to address this need by harnessing the power of deep learning techniques.

1. The Dataset YogaCE

One of the primary achievements of the thesis is the construction of the YogaCE dataset. This endeavor comprised two fundamental phases. Firstly, images were curated from pre-existing online datasets, selected, filtered, and organized to shape the foundation of the new dataset. Secondly, drawing insights from classical yoga pose literature and pertinent online resources, a set of standardized criteria for assessing each yoga pose within the dataset was meticulously formulated. Based on these assessment criteria, each image in the dataset was labeled as either a standard pose image or a non-standard pose image.

The data in YogaCE was sourced from two publicly available datasets on the data science competition website, Kaggle. One of the datasets, named the "Yoga Poses Dataset", consisting of five different yoga poses, was obtained using the API functionality from Bing by NIHARIKA PANDIT^[1]. The other dataset, known as the "Yoga Posture Dataset", containing six distinct yoga poses, was compiled and merged by Suradech Kongkiatpaiboon^[2] from three other different publicly available datasets. Notably, the two original datasets contained some images that featured multiple individuals, cartoon illustrations, incomplete depictions of body parts, or variations of yoga poses not aligned with the intended yoga posture. These images were considered erroneous, abnormal, or not in line with the scope of this

experiment. Therefore, in the selection of the dataset for this study, it was necessary to exclude this subset of data. Additionally, to mitigate class imbalance issues and enhance the generalization and stability of models, careful consideration was given to controlling the number of images for each category when selecting dataset images, to achieve a nearly balanced distribution across all categories.

The meticulous instructions and reference images documented in the work "Light on Yoga" by the renowned yoga master B.K.S. Iyengar^[3] serve as the primary criteria for determining the standard execution of yoga poses. Furthermore, guidance from professional sources found on the internet, which highlight key points and common errors associated with each yoga pose, is utilized as supplementary criteria for distinguishing the standard and non-standard execution of poses in the dataset. This approach combines the wisdom of a respected yoga master with contemporary expert insights to ensure the accuracy and reliability of assessing yoga pose images. For each of the five yoga poses included in the YogaCE dataset, the thesis outlines the stringent criteria for determining the standard execution of each pose, with the processes of deriving these criteria explained. Additionally, the thesis provides visual examples of standard pose executions, accompanied by exhaustive listings and illustrative images of the common errors associated with each pose.

The completed YogaCE dataset encompasses a total of 1,073 clear and well-composed single-person yoga pose images. It features five distinct yoga poses: Downward-Facing Dog, Tree Pose, Warrior II, Low Cobra, and Goddess Pose. Within each pose category, there are approximately 200 images, further categorized into two groups: standard pose images and non-standard pose images. The tabulated information in Table I provides a breakdown of the quantities of standard and non-standard pose images for each pose category, along with the overall image count.

Table I : Quantitative Overview of Images in YogaCE

Image Category	Downward-Facing Dog	Tree Pose	Warrior II	Low Cobra	Goddess Pose
quantities of standard pose images	131	124	83	128	92
quantities of non-standard pose images	105	82	114	112	102
overall image count	236	206	197	240	194

Based on the comprehensive literature review of existing yoga pose datasets in the thesis, it is evident that the predominant focus of current datasets centers around the task of pose classification. These datasets typically involve annotating images with specific pose categories, yet they often overlook the critical aspect of evaluating the correctness or standardization of the portrayed yoga poses. In contrast, the YogaCE dataset goes a step further by introducing classification labels for whether each yoga pose is standard or non-standard. This augmentation of classification labels based on the correctness of yoga poses provides a valuable foundational dataset for research in yoga pose assessment. It paves the way for potential breakthroughs in the development and refinement of yoga pose evaluation models.

2. Yoga Pose Analysis Based on Multi-modal Algorithms

The yoga pose analysis research in the thesis is divided into two main components: multimodal-based yoga pose recognition and multimodal-based yoga pose evaluation. Both the recognition models and the evaluation models utilize common classification model frameworks, differing primarily in the number of neurons in the output layer. Consequently, after the design and establishment of the models, experiments are conducted separately for these two components. The models take input from two modalities of data: firstly, the RGB data of images in the YogaCE dataset, and secondly, the 3D coordinate data of 33 human body keypoints, which is extracted using the MediaPipe BlazePose network^[4] (A multimedia machine learning model application framework developed and open-sourced by Google Research) from images in the YogaCE dataset. In order to obtain the optimal yoga pose analysis model through comparative experiments, the thesis introduces three multimodal fusion approaches. These approaches encompass data-layer fusion, feature-layer fusion, and decision-stage fusion.

Within the data-layer fusion approach, human body skeletal data extracted from images in the YogaCE dataset using the MediaPipe BlazePose network are superimposed onto the original images. This overlay is created by plotting the skeletal data points as keypoints and connecting them with lines, a process facilitated by MediaPipe's integrated visualization feature. The outcome consists of yoga pose images enriched with superimposed human body skeletons, seamlessly integrated into the original images. The data-layer fusion approach employs the visualized yoga pose image dataset, derived from the original dataset yogaCE, as its input data. Classification results are derived by processing this dataset through a VGG19 neural network.

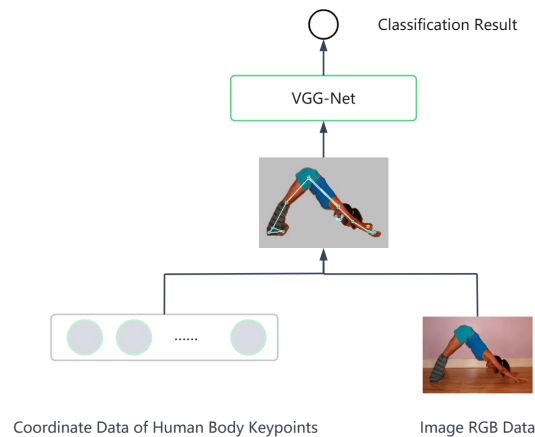


Fig.1. The model architecture diagram for the data-layer fusion approach

Within the feature-layer fusion approach, the network design consists of two branches: the MLP network branch and the VGG network branch. The MLP network branch is a three-layer MLP network with the removal of the dense layer, while the VGG network branch is based on the VGG19 network with the dense layer removed. The keypoint coordinate data extracted from images in the YogaCE dataset is fed into the MLP network branch, and the RGB data from the YogaCE dataset is fed into the

VGG network branch. These two branches are concatenated at the fully connected layer, and the combined output serves as the result, thus forming an integrated classification model.

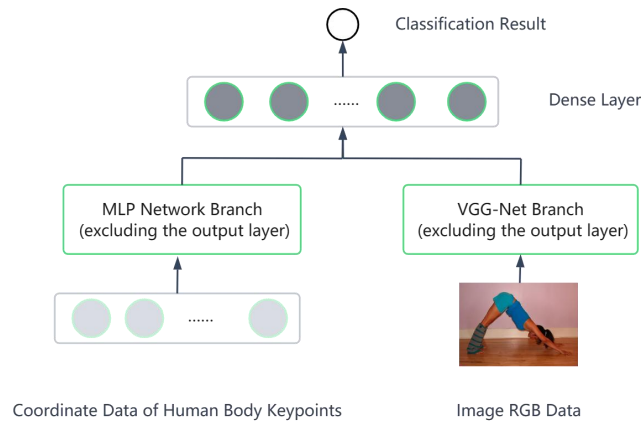


Fig.2. The model architecture diagram for the feature-layer fusion approach

Within the decision-stage fusion approach, the keypoint coordinate data extracted from images in the YogaCE dataset is fed into a three-layer MLP network, while the RGB data from the YogaCE dataset is fed into a VGG19 network, resulting in two separate classification models. The outcomes of these two models are subsequently integrated at the decision stage, employing predefined weights for combination, ultimately yielding the definitive classification result. A series of experiments are conducted with different weight combinations, with the resulting accuracies compared. The weight combination that yields the highest accuracy is chosen as the final weight parameter.

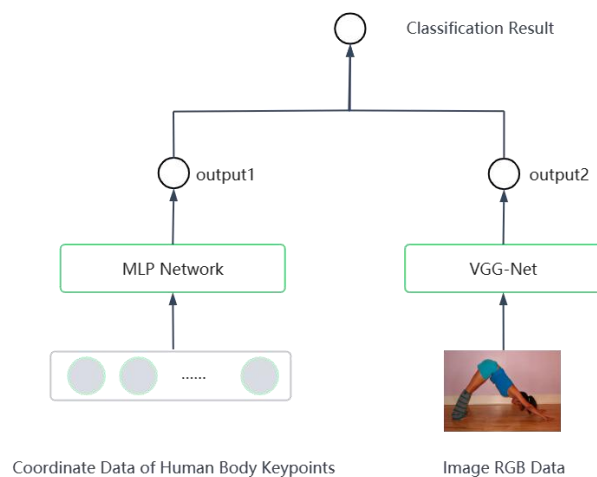


Fig.3. The model architecture diagram for the decision-stage fusion approach

The experiments followed a 7:3 split ratio for training and testing datasets, with additional preprocessing involving image augmentation applied to the RGB data. In the development phase of the

yoga pose recognition model, named YogaRegNet, five comparative experiments were conducted. These experiments included three multimodal data fusion models and two single-modal data models. The latter involved the model inputting RGB image data into a VGG19 network and the model feeding human body skeletal coordinate data into a three-layer MLP network. For the training of each model, special attention was given to ensure the convergence of the loss function, while optimizing learning rates. Within the YogaCE dataset, containing five distinct yoga pose categories, a corresponding yoga pose evaluation model named YogaEvaNet was established for each category. These evaluation models underwent the same five comparative experiments as previously detailed. The final model configuration for each net within YogaRegNet or YogaEvaNet was selected based on the experiment yielding the highest testing accuracy.

The experimental results clearly show that YogaRegNet achieves the highest accuracy, reaching an impressive 99%, when employing the data-layer fusion approach. Consequently, it was selected as the final model configuration. In the case of YogaEvaNet, which includes evaluation models corresponding to five yoga pose categories, three of these categories (Downward-Facing Dog, Goddess Pose, and Low Cobra) achieved the highest overall accuracy in the decision-stage fusion model, with percentages of 76%, 76%, and 80%, respectively. Conversely, two other categories (Tree Pose and Warrior II) obtained the highest overall accuracy in the data-layer fusion model, with percentages of 69% and 80%. Furthermore, it's worth noting that in each model's internal comparative experiments, multi-modal data fusion models consistently outperformed or matched the accuracy of their single-modal counterparts, underscoring the effectiveness of multi-modal models in yoga pose analysis tasks.

The construction of the above-mentioned network models holds a certain degree of significance. YogaRegNet demonstrates a high accuracy and advanced performance among the existing yoga pose recognition networks. YogaEvaNet complements the current deep learning-based yoga pose analysis research by focusing on yoga pose evaluation. Additionally, both of the yoga pose analysis networks have explored the integration of multi-modal data in yoga pose analysis, exhibiting superior accuracy compared to single-modal data models.

References

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基于深度学习的瑜伽动作识别与评估

摘要: 近年来,随着计算机视觉等技术的快速发展,以及人们对便捷且有效的体育运动新方式的需求增加,智能体育已经成为一个新的趋势。一方面,瑜伽动作公开数据集主要聚焦于动作分类任务,而较少关注具体的瑜伽动作是否标准。另一方面,现有的瑜伽动作智能化算法研究也主要集中于动作识别任务,而忽略了对动作质量的评估。因此,本论文将尝试从以下两方面展开研究。

第一,通过整理和标注工作完成了瑜伽动作分类和评价数据集 YogaCE。该数据集由 5 种体式类别的瑜伽动作图片组成,每种类别由标准动作图片和不标准动作图片构成。主要实现过程为收集整理网络上现有的瑜伽动作图片数据集,并通过资料搜集定义每种体式的动作标准,对图片中瑜伽动作的标准性进行标注。

第二,基于瑜伽动作识别与评估的任务,提出了图像 RGB 数据及人体骨架坐标两种模态数据融合的多模态模型。基于现有的简单 MLP 网络和经典卷积网络 VGG 网络,总共建立了数据层融合、特征层融合和决策阶段融合三种模型方案。将这些模型分别在数据集 YogaCE 上进行训练,对比实验结果,得到最佳实验结果方案下的瑜伽动作识别模型 YogaRegNet 和瑜伽动作评估模型 YogaEvaNet。

实验结果显示瑜伽动作识别通过数据层融合构建的模型识别准确性最佳,而瑜伽动作评估在下犬式、女神式、地位眼镜蛇式通过决策层融合构建的模型评估有效性最佳,在树式、战士第二式通过数据层融合构建的模型评估有效性最佳。同时,在两个互补模态数据上训练得到的模型精度普遍高于在单模态数据上训练得到的模型精度,验证了双模态模型在瑜伽动作分析任务上的有效性。

关键词: 深度学习; 瑜伽动作; 多模态; 动作识别

Recognition and evaluation of yoga poses based on deep learning

Abstract: With the rapid development of technologies like computer vision in recent years, as well as the demand for convenient and effective new ways of sports under the current fast-paced life, intelligent sports has become a new trend. On one hand, existing public datasets for yoga movements primarily focus on action classification tasks, paying less attention to the specific standardization of yoga poses. On the other hand, current research on intelligent algorithms for yoga movements mostly concentrates on action recognition tasks, neglecting the evaluation of movement quality. Therefore, the research is carried out from the following two aspects.

Firstly, through sorting and labeling, YogaCE, a data set for the classification and evaluation of yoga poses is completed. The data set consists of yoga posture pictures of 5 different poses, each consisting of standard movement pictures and non-standard movement pictures. The main realization process is to collect and sort the data set of the existing yoga poses pictures on the network, define the standard of each pose

through information collection, and mark the standardization of yoga poses in pictures.

Secondly, according to the task of yoga poses recognition and evaluation, a couple of multi-modal models which merge RGB data of images and coordinates of human skeleton is proposed. On the basis of the existing MLP network and classical convolutional network VGG-Net, three model schemes containing data layer fusion, feature layer fusion and decision stage fusion are established. In comparison of the classification results of these models trained on the YogaCE dataset, the yoga poses recognition model YogaRegNet and yoga poses evaluation models YogaEvaNet are obtained under the optimal experimental results scheme.

The results of the experiences show that the accuracy of yoga poses recognition model constructed by data layer fusion is the best. Meanwhile, for the downward-dog pose, the goddess pose and the low-cobra pose, the evaluation effectiveness of yoga poses evaluation model constructed by decision stage fusion is the best. For the tree pose and the warrior II pose, evaluation model constructed by data layer fusion is the best. At the same time, the accuracy of the model trained on the two modal data is generally higher than that trained on the single modal data, which verifies the effectiveness of the two-modal networks in the analysis of yoga movements.

Key words: Deep Learning; Yoga poses; Multi-modal; Motion Recognition

1 前言

1.1 课题背景和意义

在我国经济快速发展的今天，人们日常快节奏的生活模式和久坐不动的生活习惯带来了许多亚健康问题。党的十八大以来，围绕全民健身工作作出了一系列重要指示和重大部署，以习近平同志为核心的党中央站在时代发展的前沿，高度重视全民健身事业发展。在全民健身事业的发展实现过程中，人们在日常健身锻炼方面时间、精力的缺乏这一问题的解决将会带来很大的推进。同时，在防控新型冠状病毒疫情的时代背景下，外出身锻炼的安排常常会受到疫情管控因素的阻碍，带来疫情传播的风险。因此，居家健身在这一时代背景下成为人们健身习惯的新风口，既节约了运动健身的时间、经济成本，又解决了疫情防控带来的出行限制问题。

瑜伽是一种身心灵锻炼的修习法，目前在世界上广为流传。瑜伽运动易于修习、体力消耗适当，是一项适合大融入日常生活的体育活动。练习瑜伽对身心健康的益处颇多，包括缓解焦虑^[1]、促进积极情绪^[2]、缓解疲劳^[3]、改善身体形态^[4]、改善心血管机能^[5]。然而，不正确的练习方式可能会带来不体态的形成和受伤的风险，与增强身体健康的锻炼初衷背道而驰。练习时专业的动作评估和指导将会缓解这一问题，但请教瑜伽指导老师的金钱成本较高，需要抽出完整的时间赴约，种种限制因素致使其普及程度有限。

随着人工智能、互联网、大数据、物联网、AR/VR 等技术的发展，各种智能运动用品和服装逐渐成为人们生活中不可或缺的一部分，受到消费者的热捧，如各种智能穿戴设备、智能运动器材和设备、智能运动场地、运动 APP 等。智能体育也成为体育运动在智能化时代下的一个新趋势。而将瑜伽动作的评估和指导智能化，将有效节约人力成本，解决居家健身的条件需求，加强促进全民健身事业的发展和实现。

综上所述，本文对瑜伽动作识别评估研究的意义在于提高瑜伽动作识别的准确度和瑜伽动作评估的有效性，推进瑜伽动作评估和指导的智能化。对智能体育领域具有重要的现实意义，对智能硬件的发展有长远的应用场景。从长远来看，也对瑜伽运动的大众化普及有推动意义，促进全民健身事业的发展。

1.2 国内外研究现状

1.2.1 图像识别研究现状

图像识别是计算机视觉领域的一个重要分支，指区分不同类别的图像，其应用覆盖了生物医学、农业、自动化、军事、安全等人类社会生活的许多方面。

图像研究始于 20 世纪 40 年代，在 20 世纪 60 年代随着人工智能的出现而迅速发展。传统的图像识别方法有反向传播算法^[6]、贝叶斯分类法^[7]等，以浅层次结构模型为主，提取像素级的低级特征，需要人为对图像进行预处理，因此图像识别的准确率不够高。深度学习（Deep Learning）是一种网络结构，它包含多隐藏层和多感知能力，对物体的属性和特征描述得更加抽象和深入。将深度学习应用在图识别领域大大提高了准确率。图像识别中的常见的深度学习模型包括深层信念网络^[8]、卷积神经网络、循环神经网络^[9]生成式对抗网络^[10]、胶囊网络^[11]等。其中，卷积神经网络是完成图像识别任务的最佳

算法之一^[12]，且近几年的 ImageNet ILSVRC 图像分类算法冠军模型均是以 CNN 模型为基础，深度优化发展而来^[13]，可见卷积神经网络在图像识别方面展现出较明显的优势

2012 年，Alex Krizhevsky^[14]提出了包含 7 层隐藏层，逐层提取图像特征的 AlexNet 网络模型，ImageNet 数据集中失误率 15.4%，获得 ImageNet ILSVRC-2012 图像分类最高精度。2014 年，Szegedy C^[15]等人设计了 GoogLeNet 网络模型，提出了架构 Inception，同时分支架构也被提出。该模型将 ImageNet 数据集中地的误差率调低至 6.7%。Simonyan K 等人^[16]提出了 VGG-Net 模型，该模型使用了 19 卷积网络，同时使用了小尺度的卷积核，减少了模型参数的数量，在 ImageNet 数据集中达到了 7.3%的错误率。2015 年，何明等人^[17]提出了 ResNet 网络模型，在 ImageNet 数据集实现了 3.6%的错误率，低于人类的平均识别水平。VGG-Net、GoogLeNet 以及 ResNet 都通过加深网络层数，优化网络能，提高了模型分类结果的准确率。2019 年，Mingxing Tan 等人^[18]使用神经体系结构搜索来设计一个新的基线网络，并将其扩展以获得一系列模型，提出了 EfficientNet，其 efficientnet b7 在 ImageNet 上实现了最先进的 84.3%精度，并推断比现有最好的卷积网络小 8.4 倍，快 6.1 倍。在 CIFAR-100(91.7%)、Flowers(98.8%)和其他 3 个迁移学习数据集上也传输良好。2020 年，Boyan Zhou 等人^[19]提出了双分支神经网络，构建了两条分支，分别训练分类能力和特征提取能力模型的双分支共享参数，然后对这两个分支进行动态加权，二者协同促进深度学习在长尾分布上的泛化能力。

在图像识别这一研究领域中，细粒度图像识别是其中一项基础研究课题，旨在对某一传统语义类别下细粒度级别的不同子类类别进行的视觉识别，如不同子的狗、不同子类的鸟、不同车型的汽车等。在传统的细粒度图像识别任务中，图像中的背景噪声通过标注框来消除，以定位目标；局部区域特征提取通过位置标注来实现，这些算法过度依赖于人工标注^[20]。近几年基于 CNN 的细粒度图像识别方法^[21,22]越来越成熟，其提取的特征拥有更强大的表达能力，在细粒度图像识别任务中通常能够取得良好的结果。在 2020 年亚洲计算机视觉会议(ACCV 2020)上,进行了首届网络监督细粒度图像识别国际性挑战赛 Web FG 2020,主要解决网络监督下的细粒度图像识别问题。其中，Net Ease Games AI Lab 团队获得冠军，该队使用聚类 and 知识蒸馏对数据进行清理，使用 Efficient Net^[18]、Res Net^[17]和双分支网络^[19]作为主干网络，获得最佳测试精度 71.43%，显著高出该数据集在普通 ResNet-50 网络上的测试精度（42%）^[23]。

综上所述，随着深度学习技术的发展，在通用图像识别任务领域已经实现了较成熟的技术，分类精度已经达到或者超过人类水平。然而细粒度图像的识别有着数据集中的类别之间差异较小的特性，针对特征领域的细粒度分类问题依然面临很大挑战。

1.2.2 人体姿态估计研究现状

人体姿态估计通常是针对单个 RGB 图像，通过对人体关键点定位来估计人体的位置和关节点位置，从而从图像和视频等输入数据中构建人体表示（例如身体骨架）。它是计算机视觉中一个极具研究价值的基础问题之一，也是机器理解人类动作、行为的重要方法领域，在体育健身、智能安防、增强现实、医疗康复等方向都有丰富的应用场景。

从研究方法上看，人体姿态估计主要分为基于传统方法与基于深度学习两个大的类别。早期的传统方法主要基于人体先验几何信息以及图结构模型进行建模，需要人工设计先验和提取特征，工作量

较大。深度学习的出现为该问题的研究提供了新的思路，且逐渐成为了主流趋势。

传统的姿态估计方法主要基于图结构模型，图结构模型是由 Fischler 等人^[24]在 1973 年提出，该算法将人体和物体表示为多个相互之间具有空间约束的部件的集合，在此基础上局部匹配并用优化算法优化。2005 年，Felzenszwalb 等人^[25]将图结构模型首次应用到了人体姿态估计领域，他们将对象定义为一组可变性的部件，他们将对象定义为假设人体各部分空间约束为树结构的一组可变性组件，并使用动态规划算法求解。此后，此方法又衍生了改进算法，2008 年，Felzenszwalb 等人^[26]在之前的工作进行了优化，提出可变形部件模型（DeformablePart Model,DPM）。该模型使用新的结构来建模映射到人体结构，取得了更好的模型精度。同年，H.Jiang 等人^[27]为了解决人体姿势中的自遮挡问题，提出了非树型结构的模型，该方法将最优部位定位表示为由图模型构造的一组相关网格中的多个最短路径问题，使用线性支持向量机来学习模型中的权重和偏置。以上这些模型大部分仅对局部特征进行建模，因此表达能力有限，在实际应用中难以发挥很好的效果。

2014 年，DeepPose^[28]首次将深度学习方法引入到人体姿态估计领域，它使用 AlexNet 骨干网络提取图片特征，然后通过级联回归器得到精确的人体关键点位置。该网络在 LSP 数据集上训练，在 ImageParse dataset 上测试得到的各部位平均测试精度为 69%，超越了当时已知的一些人体姿态估计方法。2015 年，Tompson 等人^[29]提出利用 CNN 来预测热图（heat maps），将估计人体姿态的问题转化为热图回归问题，以热图编码关键坐标位置。2016 年 Wei 和 Ramakrishna 等人^[30]提出了卷积姿态机（Convolutional Pose Machines, CPM）模型来回归热图，从而定位人体关键点。该方法将卷积网络整合到姿态机器框架中，且每个阶段的网络都有监督信息，加强中间监督，从而补充反向传播的梯度并调节学习过程，解决了训练过程中梯度消失的问题。该模型的 PCKh-0.5 得分为 78.28%，该精度显著超越了论文中给出的对比模型的表现。2017 年，Z.Cao 等人^[31]提出了 OpenPose，采用自下而上的方法，先检测一幅图像中所有的关节节点坐标，再进行坐标聚类，形成与人对应的关键点坐标，是首个用于多人二维姿态检测的开源实时系统。2020 年，Valentin Bazarevsky 等人^[48]提出三维姿态检测开源实时系统 MediaPipe BlazePose，该模型使用编码器-解码器网络结构来预测所有关节的热图，采用热图、偏移和回归相结合的方法进行关节位置的预测。BlazePose 分为完整版 BlazePose Full 与轻量化版 BlazePose Lite，与当时较为先进的 OpenPose 相比，BlazePose Full 在正在进行健身或瑜伽人群为图片主体的数据集 Yoga Dataset, PCK@0.2 上的精度为 84.5%，高于 OpenPose 测试得出的精度 83.4%，且 BlazePose Full 在单个中间层手机 CPU 上的运行速度比 OpenPose 在 20 核桌面 CPU 上的运行速度快 25-75 倍，在检测运动健身人群的人体姿态和运行速度上展现出了其优势。

综合来看，人体姿态估计在其发展历程中已经取得了较大的进展，但准确识别依然有一定的难度，未来仍有较大的发展提升空间。

1.2.3 瑜伽动作数据集研究现状

现有的公开瑜伽动作数据集主要由由图片构成的数据集。其中所包含的瑜伽体式类别数量各不相同。

2020 年，Verma M 等人^[32]提出数据集 Yoga-82，该数据集由 82 个类别的瑜伽动作图片组成，每个图片标注了三个层次的标签，其中第一层为身体姿势的大类（例如站立、坐下、倒立等），第二层为

身体姿态的变化，第三层为具体的体式名称。其数据来源是 3 个提供了瑜伽动作照片的网站。同年，数据竞赛网站 Kaggle 的用户 NIHARIKA PANDIT^[51]上传了数据集 Yoga Poses Dataset，该数据集由 5 种瑜伽体式的瑜伽动作相片组成，这些图像通过在网络上搜索关键字爬虫获取，因此包含了现实拍摄相片和绘画作品。图片总数共 1551 张，每张图片标注了其体式类别。2021 年，Laurence Moroney^[33]在其网站上传了一个未命名的瑜伽动作数据集，该数据集由 5 种瑜伽体式的瑜伽动作图片组成，这些图片通过 Daz3D 建模创建，包含了数百张 300×300 的彩色图像，每张图像有不同的背，景涵盖多种肤色、性别、头发颜色，模拟了不同的相机角度和灯光，每张图片标注了其体式类别。2022 年，Kaggle 用户 SURADECH KONGKIATPAIBOON AND 1 COLLABORATOR^[52]上传了数据集 Yoga Posture Dataset，该数据集的来源即前述的现有 3 个经典的瑜伽动作数据集，经过作者的整理，共合并为 6 种瑜伽动作体式，每张图片标注了其体式类别。

现有的瑜伽动作数据集各有其特点，例如涵盖瑜伽体式多样，数据来源和图片形式独特等，但整体来看基本都聚焦在动作分类任务中，仅仅标注了图片种动作的动作或体式类别，而较少关注具体的瑜伽动作是否标准，使其应用领域比较局限，难以在瑜伽动作质量评估的研究中使用。

1.2.4 瑜伽动作的识别与评估研究现状

瑜伽动作识别是人体动作识别在瑜伽运动领域的一项研究，针对从图像或视频数据中提取的特征进行分类任务，判断出其中所包含的动作。动作识别根据从数据中获取人体动作方式的不同，可分为基于 RGB 图像、基于骨骼、基于传感器数据等类别。而瑜伽动作评估则是人体动作评估在瑜伽运动领域的一项研究，通过对图像或数据进行分析，对其中的人体动作给出质量评估。动作评估的主要研究路线集中在动作细粒度分类^[34]，基于人体重构和人体姿态估计^[35]，事件发现任务^[36]，基于语义信息评估^[37]等几个方向。

Hua-Tsung Chen 等人^[38]基于 Kinect 对人体形态进行抓取，将目标物质心与轮廓极值之间的快速骨架化技术——星形骨架作为人体姿态的代表描述符用于瑜伽动作识别。张梦莹^[39]通过 Kinect 收集标准瑜伽动作和训练师的瑜伽动作数据，利用动态时间规整算法和 Hausdorf 距离算法融合的方法，以阈值识别动作名称，并以关节角度测量的动作评估算法，对动作不到位的关节进行评估，从而得出动作的相似度。王芬^[40]利用 Mask R-CNN 改进的瑜伽动作图像数据的动作识别，将原来的标准卷积替换为深度可分离卷积，通过对分枝中卷积部分的改进来提高网络效率。万益^[41]将 RGB 图像数据和骨骼数据输入联合模型中，联合模型将输出瑜伽动作的类别以及这种动作的等地评分，通过多模态数据完成瑜伽动作的识别及评估。Madhura Praksh 等人^[42]采用 ml5.js、PoseNet 和神经网络技术，实时采集视频输入，进行瑜伽姿势的分类。Ze Wu 等人^[43]采用反向传播人工神经网络作为第一分类器及采用模糊 C-means 作为第二分类器对瑜伽姿势进行分类，并将身体各部位的姿态数据作为多维高斯变量，构建贝叶斯网络。以各身体部位对应高斯变量相对于连通身体部位对应高斯变量的条件概率为标准，定量评价身体部位的标准度，对身体部位作出动作指导。

在研究方法上，现有的瑜伽动作的智能化分析多数集中在动作识别任务，而关注到动作质量评估的研究相对较少，然而在实际应用中动作质量的评估更具价值，可以智能化地减轻不标准的动作在运动过程中带来的健康风险。在研究的数据选择上，现有方法多数仅应用了单一的 RGB 或骨架数据，无

法根据两个模态的互补信息获得更好的特征表示，因此仅能获得次优的分类性能。

1.3 研究内容及论文结构

1.3.1 研究内容

本文的任务主要是基于深度学习的图像识别算法及多模态网络，实现对瑜伽动作图像的识别与评估。主要研究内容如下：

（1）整理标注瑜伽动作分类评价数据集

整合网络上现有的瑜伽动作图像数据集，基于现有的权威瑜伽体式动作标准，对每张瑜伽动作图像进行标准与否的标准，最终确保各瑜伽动作类别的图片数量相近，且各瑜伽动作类别中的标准动作图像与其中的不标准动作图像的图片数量相近，在这一标准的基础上筛选数据集中的图像，整理为一个完整的瑜伽动作分类评价数据集。

（2）构建瑜伽动作分类与评估网络

研究人体姿态估计中常用的人体骨架提取框架，选取使用于运动分析的网络对瑜伽动作图片进行骨架数据的提取。研究图像识别中常见的网络，分析其特点并对比后，参考设计瑜伽动作分类与评估使用到的主干网络。研究现有的常见多模态数据融合方法，设计几种多模态融合网络。将整理标注完成的瑜伽动作分类评价数据集在这几种多模态融合网络上展开实验验证，通过构建评价标准并对比选取最佳的模型策略，确定选用的网络模型。

具体研究对象及研究方法如下。

1.3.1.1 研究对象

本文以数据科学竞赛网站 Kaggle 上的两个公开数据集 Yoga Poses Dataset^[50]和 Yoga Posture Dataset^[51]为数据来源的自整理、标注数据集 YogaCE 作为研究对象。

1.3.1.2 研究方法

1 文献资料法

查阅各种书刊、杂志、专著与本研究有关的书籍，通过收集瑜伽动作各体式标准动作等方面的研究成果，进行整理归纳。在互联网通过知网 CNKI、谷歌学术检索有关图像识别、人体姿态估计及瑜伽动作识别和评估有关的文献，检索词为“图像识别，人体姿态估计，瑜伽动作”。

2 实验法

在瑜伽动作识别与评估的模型建立过程中，将 RGB 图像数据与人体骨架数据两种模态数据的融合分为前融合、神经网络阶段后融合以及决策阶段后融合三种实验方案，分别用实验数据集进行模型的训练，并进行测试结果精度的对比。

3 计算机程序设计法

本研究计划使用 Python 作为编程语言，在人体骨架数据的提取方面用到 MediaPipe 模块给出的骨架提取用法，在深度学习模型方面基于 PyTorch 框架进行构建，实现网络的训练和测试。

1.3.2 论文结构

本文分为五章内容：

第一章：前言。简要介绍本文的研究背景和研究意义，图像识别、人体姿态估计、瑜伽动作识别与评估的国内外研究进展，同时阐述本文的主要研究内容。

第二章：相关基本技术概述。本章介绍瑜伽动作分析网络中神经网络的基本技术和人体骨架提取框架 MediaPipe BlazePose 基本技术。神经网络技术包括瑜伽动作分析网络使用到的卷积神经网络和 MLP 网络的基本结构以及神经网络的训练过程，MediaPipe BlazePose 技术包括该框架的基本原理、神经网络结构和骨架提取结果展示。

第三章：瑜伽动作分类评价数据集 YogaCE。本章介绍了自整理标注的瑜伽动作分类评价数据集 YogaCE 的数据来源与选取规则，并给出最终得到的数据集的各类别图片数量。引用权威文献中的动作标准论述其图像标注判断依据。

第四章：基于多模态的瑜伽动作分析。本章介绍了瑜伽动作分析实验中使用到的 3 种多模态融合方案，及其中设计到的 3 种主干网络结构。同时说明了数据集准备、实验设置及结果的评价标准。最后给出实验结果，对实验结果进行分析，并讨论实验中的参数选取确认。

第五章：总结与展望。简要总结本文的研究和实践内容，阐述研究内容存在的不足之处，展望未来研究工作。

2 相关基本技术概述

2.1 神经网络

2.1.1 卷积神经网络基本结构

卷积神经网络^[44]（Convolutional Neural Network，简称 CNN）是一种深度学习模型，广泛应用于图像识别、计算机视觉等领域。其基本结构包括卷积层、池化层、激活函数、全连接层和输出层等。在实际应用中，卷积神经网络通常由多个卷积层、池化层和全连接层交替堆叠组成，每层的输出作为下一层的输入，构成一个端到端的特征提取和分类模型。其所包含的各结构内容具体介绍如下。

2.1.1.1 卷积层

卷积层是 CNN 最重要的部分，通过卷积核对输入数据进行特征提取和转换。卷积层的核心操作是卷积运算，即将一个卷积核应用于输入图像的每一个像素，得到一个输出矩阵。卷积核是一个小的二维矩阵，其中的每一个元素称为卷积核权重，它通过滑动窗口的方式在输入图像上移动，对每一个窗口内的像素进行加权求和，得到一个输出值。将所有窗口的输出值组合在一起，即可得到输出矩阵。

2.1.1.2 池化层

池化层用于缩小卷积层的输出尺寸，减少模型复杂度并提高模型的鲁棒性。池化操作可以采用最大池化或平均池化等方式，将特征图中每个小块的最大值或平均值作为输出，从而减小特征图的大小。

2.1.1.3 全连接层

全连接层用于将卷积层和池化层的输出进行展开，并连接到输出层进行分类或回归等任务。全连接层的每个神经元都与前一层的每个神经元相连，需要进行大量的计算和参数调整。

2.1.1.4 激活函数

激活函数用于对卷积层的输出进行非线性变换，引入非线性因素。常用的激活函数包括 ReLU、Sigmoid、Softmax 和 Tanh 等。

ReLU 函数是一种常用的非线性激活函数，通常用于神经网络中的隐藏层，优点是计算简单，可以解决梯度消失问题。其计算为在输入为正数时返回该值，而在输入为负数时返回 0。其公式为：

$$\text{ReLU}(x) = \max(0, x) \quad (2-1)$$

Sigmoid 常被用于解决二分类问题，其计算为将输入映射到一个介于 0 和 1 之间的输出。其公式为：

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (2-2)$$

Softmax 函数是一个用于将输入转换为概率分布的函数。它可以将任何实数向量映射到一个介于 0 和 1 之间的概率分布，使得各个元素之和等于 1。在神经网络中，Softmax 通常被用于分类任务中，将神经网络的输出转化为类别概率分布。其公式为：

$$\sigma(x) = \frac{e^{z_i}}{\sum_{k=1}^k e^{z_i}} \quad (2-3)$$

Tanh 函数是一种常用的双曲正切函数，它是一种 S 形曲线，其值介于 -1 和 1 之间。与 Sigmoid 函数类似，Tanh 函数也是一个平滑且可导的非线性函数。比较常用在处理时序数据。其公式为：

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (2-4)$$

2.1.2 MLP 神经网络基本结构

MLP (Multi-Layer Perceptron) 神经网络^[44]是一种常见的前向人工神经网络，广泛应用于图像识别、自然语言处理、音频识别、推荐系统和时间序列预测等任务中，由于其广泛的应用性和强大的学习能力，已成为现代深度学习中最基础和最重要的一部分。MLP 网络由多个神经元层组成，其中第一层是输入层，最后一层是输出层，中间的层被称为隐藏层。每个神经元都接收来自前一层的输入，计算加权和并最后通过激活函数输出结果。

2.1.2.1 输入层

输入层通常是神经网络的第一层，接收外部输入并将其传递到下一层。输入层的神经元数量由输入特征的数量决定，每个神经元对应一个输入特征。例如，对于一张图片，每个像素的强度可以作为一个输入特征，那么输入层的神经元数量就是像素数目。

在输入层中，通常不需要进行任何处理或转换。输入数据可能需要被标准化或缩放，以确保输入

数据的值域在相同范围内，以避免特征之间的偏差。

2.1.2.2 隐藏层

隐藏层是神经网络中的中间层，其神经元数量可以根据具体问题和模型需求进行调整。隐藏层的作用是将输入数据进行非线性转换，以捕捉输入数据中的更高级别的特征。这种转换是通过权重和偏置的矩阵乘法和激活函数的组合实现的。

隐藏层可以有多个，每个隐藏层都会计算出一个新的特征表示。这些新的特征表示是根据输入数据中较高级别的模式和特征来计算的，由此更好地捕捉到数据的内在结构和关系。

2.1.2.3 输出层

输出层通常是神经网络的最后一层。该层根据模型的任务类型，可以有一个或多个神经元。例如，对于二分类问题，输出层只需要一个神经元，输出 0 或 1；对于多分类问题，输出层需要多个神经元，每个神经元表示一个类别。

输出层的神经元通过权重和偏置的矩阵乘法和激活函数来计算输出。通常使用不同的激活函数来处理不同类型的输出问题，例如 **sigmoid** 函数用于二分类问题，**softmax** 函数用于多分类问题。

2.1.3 神经网络的训练

神经网络的训练过程^[44]可以概括为输入数据，前向传播，计算误差，反向传播，权重更新等几个主要步骤。神经网络的训练过程是一个迭代的过程，需要不断重复执行这些步骤，直到达到停止条件。训练过程中，需要对训练集和验证集的误差进行监控，以评估神经网络的性能并及时调整训练参数。

2.1.3.1 前向传播

在前向传播过程中，输入数据通过神经网络的各层进行前向传播，最终得到输出结果。前向传播的过程中，每一层的神经元会对上一层的输出进行加权求和，并通过激活函数进行处理。该过程可以表示为：

$$a^{(l)} = w^{(l)}z^{(l-1)} + b^{(l)} \quad (2-5)$$

$$z^{(l)} = g(a^{(l)}) \quad (2-6)$$

其中， $w^{(l)}$ 表示第 l 层的权重， $b^{(l)}$ 表示第 l 层的偏置， $z^{(l-1)}$ 表示上一层的输出， $a^{(l)}$ 表示第 l 层的输入， $z^{(l)}$ 表示第 l 层的输出， $g(x)$ 表示激活函数。在前向传播过程中，神经网络会不断学习特征，并将这些特征传递给下一层网络。最终，神经网络的输出结果将用于计算误差。

2.1.3.2 计算误差

计算误差是神经网络训练的一个关键步骤。通常使用损失函数来衡量神经网络的输出结果与实际标签之间的差异。常见的损失函数包括均方误差（MSE）、交叉熵（cross-entropy）等。均方误差函数通常用于回归问题，交叉熵函数则通常用于分类问题。均方误差的计算公式为：

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2-7)$$

其中， n 表示样本数量， y_i 表示第 i 个样本的实际标签， $\hat{y}_i$ 表示神经网络对第 i 个样本的输出结果。在神经网络的训练过程中，通常使用梯度下降算法来最小化 MSE，从而使得神经网络能够逐渐学习到有用的特征。交叉熵的计算公式为：

$$CE = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log \hat{y}_{ij} \quad (2-8)$$

其中， N 表示样本数量， M 表示分类数。 y_{ij} 表示第 i 个样本的第 j 类标签，通常为 0 或 1。 $\hat{y}_{ij}$ 表示神经网络输出的第 i 个样本在第 j 类上的预测概率。交叉熵损失函数的本质是用来衡量两个概率分布之间的相似度，通过最小化交叉熵损失函数，可以使得神经网络在分类任务中得到更好的表现。误差函数的值越小，表示神经网络的输出结果越接近实际标签。在计算误差时，需要将神经网络的输出结果和实际标签进行比较，并计算误差大小。

2.1.3.3 反向传播

反向传播算法是神经网络训练的核心算法之一。在反向传播过程中，根据损失函数的大小，将误差从输出层向输入层进行反向传播，然后根据误差大小对神经网络的权重和偏置进行调整，从而使得神经网络能够逐渐学习到有用的特征。该过程首先计算输出层的误差，即计算损失函数对输出结果的偏导数；接着通过链式法则，将输出层的误差传递到隐藏层，计算隐藏层的误差假设第 l 层的第 i 个神经元的误差为 $e_i^{(l)}$ ，则有：

$$e_i^{(l)} = \frac{\partial L}{\partial z_i^{(l)}} = \sum_j \frac{\partial L}{\partial z_j^{(l+1)}} \frac{\partial z_j^{(l+1)}}{\partial z_i^{(l)}} = \sum_j e_j^{(l+1)} \frac{\partial z_j^{(l+1)}}{\partial z_i^{(l)}} \quad (2-9)$$

其中， L 表示损失函数， $z_i^{(l)}$ 表示第 l 层第 i 个神经元的加权输入， $e_j^{(l+1)}$ 表示第 $l+1$ 层第 j 个神经元的误差， $\partial z_j^{(l+1)} / \partial z_i^{(l)}$ 表示第 $l+1$ 层第 j 个神经元的加权输入对第 l 层第 i 个神经元的加权输入的偏导数。接着计算每个参数对误差的贡献，对于第 l 层的第 i 个神经元到第 $l+1$ 层的第 j 个神经元之间的权重，其贡献为：

$$\frac{\partial L}{\partial w_{ij}^{(l)}} = \frac{\partial L}{\partial z_j^{(l+1)}} \frac{\partial z_j^{(l+1)}}{\partial w_{ij}^{(l)}} \quad (2-10)$$

其中， $\partial L / \partial z_j^{(l+1)}$ 是第 $l+1$ 层第 j 个神经元输出的误差， $\partial z_j^{(l+1)} / \partial w_{ij}^{(l)}$ 是第 l 层第 i 个神经元输入到第 $l+1$ 层第 j 个神经元的权重。

最后通过误差反向传播更新权重和偏置。具体来说，对于第 l 层的第 i 个神经元第 $l+1$ 层的第 j 个神经元之间的权重 $w_{ij}^{(l)}$ ，其更新公式为：

$$w_{ij}^{(l)} \leftarrow w_{ij}^{(l)} - \eta \frac{\partial L}{\partial w_{ij}^{(l)}} \quad (2-11)$$

其中， η 表示学习率。

对于第 l 层的第 i 个神经元第 $l+1$ 层的第 j 个神经元之间的偏置 $b_{ij}^{(l+1)}$ ，其更新公式为：

$$b_{ij}^{(l+1)} \leftarrow b_{ij}^{(l+1)} - \eta \frac{\partial L}{\partial b_{ij}^{(l+1)}} \quad (2-12)$$

2.2 基于 MediaPipe BlazePose 的人体关节点提取

MediaPipe^[46]是一款由 Google Research 开发并开源的多媒体机器学习模型应用框架。其核心框架由 C++ 实现，并提供 Java、Python 以及 Objective C 等语言的支持。其中应用于人体姿态估计的版块，BlazePose^[47]，由团队在 CVPR 2020 的 CV4ARVR 研讨会上展示。BlazePose 使用机器学习从画面中含有人体的视频的一帧或一张图片中推断出人体的 33 个人体关节点所在的坐标位置，从而获取其中的人体骨架结构。与目前基于标准 COCO 拓扑结的姿势模型相比，BlazePose 可以精确定位更多的关键点，因此具有适合应用于运动健身领域的人体关节点提取的特点，是当前用于人体姿态估计的主流框架之一。

2.2.1 人体骨骼关节点提取

2.2.1.1 人体检测

大多数现代目标检测解决方案依赖于非最大抑制(NMS)算法的最后一个后处理步骤，但这种方法在检测人体时，在一些人体关节活动丰富的场景下往往出现多个不明确的框满足 NMS 算法的 IU 交集阈值，因此出现失效的问题。为了解决这一问题，BlazePose^[47]的研发者决定转化思路。经研究发现，在多数情况下，神经网络收到的关于人体位置的最强信号是人的脸。因此，BlazePose 模型用到的人体检测器以在 BlazeFace^[48]中用到的一款快速人脸检测器作为代理，先在图像中检测到人脸所在区域，即图 2-1 中的方形框所检测到的区域。此外，该检测器还可以检测到额外的人体点位，该检测过程将人体的臀部中点作为圆心，足底到臀部中点之间的距离为半径，确认出整个人周围圆圈的大小，将人体的中心、旋转、比例描述、类比为圆。因此，检测器预测出的个人特定的对齐参数为：人臀部的中点，整个人周围圆圈的大小，以及倾斜度（即两个肩膀中间点和臀部中间点的连线线之间的夹角）。图 2-1 中绿色圆形展示了人体通过检测器对齐的方位状态。

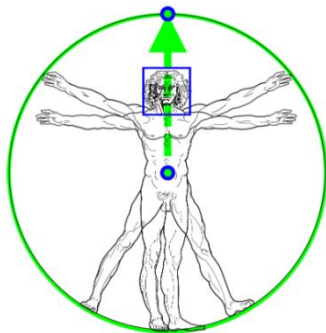


图 2-1 经检测器对齐的人体及检测框所在区域示意图

2.2.1.2 人体关键点的拓扑结构

模型中的人体提取关键点拓扑结构融合了 BlazeFace^[48]、BlazePalm^[49]及 Coco^[50]中人体关键点的一些重要点位，共选取了人体上的 33 个点，这些关键点之间通过连线构成一个形容人体骨架的拓扑结构。结构图 2-2 如所示，图中标号为 0-32 的点对应的人体部位点依次为：鼻子、左眼内侧、左眼中心、左眼外侧、右眼内侧、右眼中心、右眼外侧、左侧面部边缘、右侧面部边缘、嘴唇左侧、嘴唇右侧、左肩外侧点、右肩外侧点、左手肘、右手肘、左手腕、右手腕、左指第一关节、右小指第一关节、左食指第一关节、右食指第一关节、左大拇指第二关节、右大拇指第二关节、左髌骨、右髌骨、左膝盖、右膝盖、左脚踝、右脚踝、左脚跟、右脚跟、左第二脚趾、右第二脚趾。

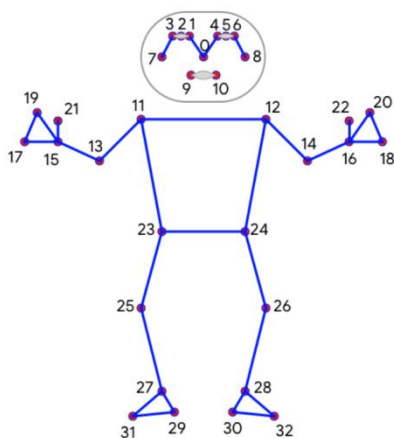


图 2-2 关键点拓扑结构

2.2.1.3 神经网络结构

模型在神经网络结构中采用热图、偏移和回归相结合的方法。模型在训练阶段用到热图和补偿损失，在这一过程中使用编码器-解码器网络结构来预测所有关节的热图，并在进行结果推断之前从模型中删除相应的输出层，从而提高网络的效率，然后再叠加使用另一种直接回归到所有关节坐标的编码

器网络。该网络输入为一张 RGB 图片，输出为上述 33 个人体关键点的三维坐标位置。其结构如图 2-3 所示。

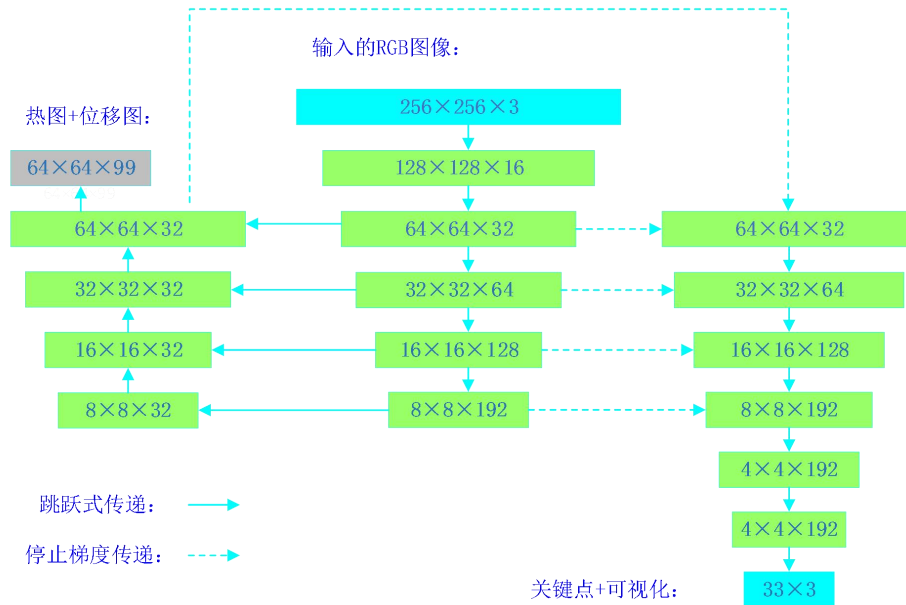


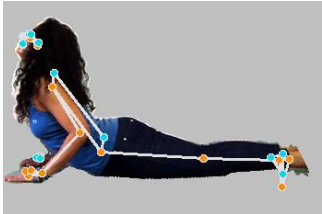
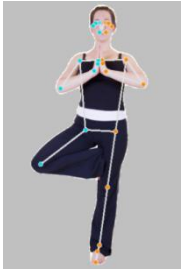
图 2-3 人体关节点提取模型网络结构

2.2.2 结果展示

在使用到的数据集中 1073 张图片中，使用 MediaPipe BlazePose 成功识别到人体的图片为 1072 张，识别率 99.91%。在成功识别到人体的图片中，识别效果在不同图片上有着不同表现，主要分为骨架提取整且基本准确，骨架提取完整但有较明显的偏差，以及骨架提取在部分人体部位上偏差过于严重以至于在可视化图片上无法得到完整骨架，这几种情况。

综合来看，骨架提取完整且基本准确的情况占大多数，在这些图片上 MediaPipe BlazePose 表现出了较好的性能。对于骨架提取完整但有较明显的偏差的情况，集中表现在脚部的点位识别有明显偏差，造成提取到的骨架中脚的点所在位置与原图片中不一致、或脚底与地面之间的夹角有明显区别。这一问题可能会给对脚部位置、姿势有标准要求的体式在评估模型的建立中造成一定干扰。对于在可视化图片上无法得到完整骨架的情况，下犬式和低位眼镜蛇式中脚部、腿部、手部的点位都有出现过于严重的偏差，战士第二式在脚部、腿部的点位有出现过于严重的偏差，女神式在脚部的点位有出现过于严重的偏差，树式由于抬起脚放置在另一大腿上，抬起脚点位的识别严重偏差情况非常多，同时也有少数手部的点位出现过于严重的偏差。每一种结果的情况及其具体问题对应的图片示例如表 2-1 所示。

表 2-1 骨架提取的不同结果情况及图片示例

结果类别	具体问题	图片示例 1	图片示例 2
骨架提取完整且基本准确	-		
骨架提取完整但有较明显的偏差	脚部的点位识别有较明显偏差		
	腿部的点位识别偏差过于严重		
在可视化图片上无法得到完整骨架	脚部的点位识别偏差过于严重		
	手部的点位识别偏差过于严重		

3 瑜伽动作分类评价数据集 YogaCE

实验用到的数据集为自整理标注数据集 YogaCE（Yoga posture Classification and Estimation dataset），采用拍摄完整、清晰的单人瑜伽动作图片，有下犬式、树式、战士二式、低位眼镜蛇式及女神式共 5 类体式，每种体式中有各 100 张左右的标准动作图片和不标准动作图片两种类别，数据集各体式类别中的标准动作图片数量、不标准动作图片数量及图片总数如表 3-1 所示。数据集准备过程中除了数据的采集整理，还对图片中动作的标准与否进行了标注。

表 3-1 各体式类别中的标准动作图片数量、不标准动作图片数量及图片总数

图片类型	下犬式	树式	战士二式	低位眼镜蛇式	女神式
标准动作图片数量	131	124	83	128	92
不标准动作图片数量	105	82	114	112	102
图片总数	236	206	197	240	194

3.1 数据来源及选取

数据的来源为数据科学竞赛网站 Kaggle 上的两个公开数据集，其中一个为由 NIHARIKA PANDIT^[51]使用 API 功能从 bing 提取，获得的包含五种瑜伽动作，命名为 Yoga Poses Dataset 的数据集，另一个为由 Suradech Kongkiatpaiboon^[52]从三个不同的公开数据集中整理合并，获得的包含六种瑜伽动作，命名为 Yoga Posture Dataset 的数据集。

这两个原数据集中各个类别的瑜伽动作图片中包含一些图片为图中有多人图片、卡通插画图片、拍摄人物展示出的肢体部位不完整或做出的动作为该瑜伽体式的变体的图片，属于错误、异常的数据或与本实验应用场景不符的数据。因此在本实验数据集的选取中需要将这一部分进行筛查，将卡通插画图片、变体动作图片、人像全身不完整图片筛除，将多人图片选择截取为单人图片或筛除。同时，为了使每个体式中的标准动作图片与不标准动作图片两个类别的数据量统一、差别不大，不需要用到两个数据集中所有正常、符合条件的数据。

战士第二式的数据选取方式为筛除 Yoga Poses Dataset 这一数据集中该体式类别里不符合条件的图片，将符合条件的图片作为实验的数据集内容。低位眼镜蛇式的数据选取方式为筛除 Yoga Poseture Dataset 这一数据集中该体式类别里不符合条件的图片，将符合条件的图片作为实验的数据集内容。其中，在 Yoga Poseture Dataset 中的眼镜蛇式这一类别包含了高位眼镜蛇式和低位眼镜蛇式。这两个细分体式在判断动作标准与否的判别标准上十分不同，如果分在一个类中别，会给瑜伽动作的质量评估的模型训练造成困难。因此，在 YogaCE 中只选取了其中的低位眼镜蛇图片。下犬式、树式的数据选取方式为筛除 Yoga Poses Dataset 这一数据集中该体式类别里不符合条件的图片作为主要数据，并在

Yoga Postr Dataset 这一数据集中该体式类别里符合条件的图片中找出不标准动作图片补充进去，合并为实验的数据集内容。女神式的数据选取方式为 YogaPoses Dataset 和 Yoga Poseture Dataset 两个数据集中的所有女神式图片，筛除不符合条件的图片后合并获得。

Yoga Poses Dataset 与 Yoga Posture Dataset 包含瑜伽体式类别的并集共 8 种，这 8 种体式各自及所有体式总数在 Yoga Poses Dataset 中的数量、在 Yoga Posture Dataset 中的数量、数据集 YogaCE 从 Yoga Poses Dataset 中获取的数量、数据集 YogaCE 从 Yoga Poseture Dataset 中获取的数量以及在 YogaCE 中的数量统计如表 3-2 所示。

表 3-2 各数据集中不同体式图片数及 YogaCE 中图片来源统计

数据集及选取情况	下犬式	女神式	平板式	树式	战士第二式	椅子式	眼镜蛇式	战士第一式	所有体式总数
Yoga Poses Dataset	320	260	381	229	361	0	0	0	1551
Yoga Posture Dataset	430	209	0	362	0	209	643	13	2036
YogaCE 从 Yoga Poses Dataset 中获取的数量	199	124	0	135	197	0	0	0	655
YogaCE 从 Yoga Poseture Dataset 中获取的数量	37	70	0	71	0	0	240	0	418
YogaCE	236	194	0	206	197	0	240	0	1073

3.2 图像标注判断依据

BK·S·艾扬格是一位印度瑜伽大师，是艾扬格瑜伽的创始人，被认为是全球最负盛名的一位瑜伽教师^[53]。其著作《瑜伽之光》^[54]是艾扬格根据自己 27 年的世界多地的瑜伽教学经验，详尽说明艾扬格瑜伽中的各种体式及呼吸控制法写成的书。全书包括 200 个体式的详细技巧并配有 592 副图片，以帮助练习者掌握相关的体式。该书是瑜伽发展历史上的重要著作之一^[55]。

在瑜伽动作图片的标准动作图片和不标准动作图片的类别标注上，就以《瑜伽之光》中各个体式的详细技巧和参考图片，作为标准动作的参照标准，同时，依据网络上较为专业的一些讲解瑜伽动作的文章中提及的体式的注意事项、典型错误，作为图片中动作的标准与否区分的辅助判断依据。在进行图像标注时，将动作完全符合《瑜伽之光》中该体式的关键动作要求的瑜伽动作图片，标注为标准动作图片；将动作和《瑜伽之光》中该体式的关键动作要求不一致的瑜伽动作图片，标注为不标准动作图片，且这些图片中的瑜伽动作往往符合网络上的相关文章提及的典型错误。但对于《瑜伽之光》中没有收录的瑜伽动作，则结合多篇文章，总结共同要点，综合分析确认标准。表 3-3 给出了各个体

式的标准动作要点以及典型标准动作图片。

表 3-3 各个体式标准动作要点以及标准动作图片

动作名称	标准动作要点	《瑜伽之光》示范图片	标准动作图片示例
下犬式	①手臂肘部伸直；②伸展背部，不拱起；③腿部绷直，膝盖不弯曲；④脚后跟和脚底完全放在地面上，双脚平行，脚趾朝向前方		
树式	①抬起腿的脚跟放在站立腿的根部，脚掌放在站立腿的大腿上，脚趾朝下；②髋部打开，脊柱挺直；③躯干背部伸直，向上伸展		
战士第二式	①双腿分开 120 到 135 厘米，右脚右转 90 度，左脚稍向右转；②左腿伸直膝盖绷直，完全拉伸左腿后部肌肉；③右大腿与地面平行，右胫骨与地板垂直；④两臂侧平举与肩齐，手掌朝下；⑤两肩基本平行于地面，脊柱挺直，上身躯干垂直		
低位眼镜蛇式	①伸直双腿，双脚相靠，膝盖绷直，脚趾向后，脚尖放在地面；②手掌不向前超过肩膀，肘部向后伸出；③躯干抬起至接近挺直状态，触底位置在盆骨附近		

动作名称	标准动作要点	《瑜伽之光》示范图片	标准动作图片示例
女神式	双脚向外张开约 45 度；②下蹲至大腿与地面平行或接近于平行状态，膝盖弯曲朝向脚尖方向，向外伸出距离超过脚踝上方；③脊柱挺直、伸长，垂直于地面	—	

针对每个体式，标准动作的标准的具体论述及对不标准动作的具体类型的列举如下。

3.2.1 下犬式

根据《瑜伽之光》中的体式讲解，艾扬格认为下犬式保持状态的标准动作应当手臂肘部伸直，头部向身体内侧并朝脚的方向，头顶着地，伸展背部，腿部绷直，膝盖不弯曲，脚后跟和脚底完全放在地面上，双脚平行，脚趾朝向前方。百度百科中下犬式的词条内容^[56]提及下犬式练习时的错误在于膝盖弯曲、脚跟踮起、背部拱起没有平直延伸。这样的姿势虽然缓解了背部和腿部的紧张感，但同时也让身体的下背部和腿部没有得到拉伸，给双肩和双手掌带来较大的压力，极易扭伤手腕关节。网站 Yoga society 中的一篇介绍下犬式相关知识的文章^[57]认为在保持下犬式动作时头部不应该垂下，而应该保持固定在两臂之间，而根据实验数据集中的情况，大多数图片的头部摆放都是保持于两臂之间，这与艾扬格提出的头顶顶底的标准相悖。针对这一矛盾，实验标注的判断不把头部的位置视作决定标准与否的关键部位，头部顶地或是保持于两臂之间都视作符合标准。

因此，在下犬式图片的图像标注过程中，将所有姿势关键点（头部放置位置除外）与《瑜伽之光》中提及的要点及示范图片（见表 3-3）相符合的瑜伽动作图像标注为标准动作图片，而出现典型错误，即脚跟踮起、膝盖弯曲、背部拱起及手臂肘部弯曲这几种情况的图片，标注为不标准动作图片。各类型典型错误及示例图片对照如表 3-4 所示。

表 3-4 下犬式各类型典型错误及示例图片

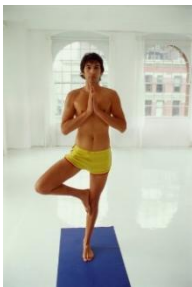
错误类型	脚跟踮起	膝盖弯曲	背部拱起	手臂肘部弯曲
示例图片				

3.2.2 树式

根据《瑜伽之光》中的体式讲解，艾扬格认为树式应当将一只抬起的脚脚跟放在另一只站立的腿的根部，脚掌放在站立腿的大腿上，脚趾朝下；根据《瑜伽之光》中给出的示例图片（图 4-2 左上图），艾扬格示范的树式背部竖直挺立、向上伸展，中心基本处于躯干的中线上。百度百科中树式的词条内容^[58]提及练习树式时容易出现的问题是抬高的那条无法打开髋部，脊柱弯曲，这些都有可能使人失去身体的平衡，受到伤害。网站 Yoga Journal 中一篇讲解联系树式的要领的文章^[59]认为练习树式时应该将脚放在站立腿内侧除膝盖以外的任何位置。压到膝盖会使关节和姿势不稳定。在《瑜伽之光》中，艾扬格提及树式练习的功效为增强腿部肌肉和平衡感，综合来看手部的摆放位置和动作并不是该体式练习的核心要求，因此在判断图片中动作标准与否时不对手的摆放作限制。

在树式图片的图像标注过程中，将所有姿势关键点（手的摆放位置和动作除外）与《瑜伽之光》中提及的要点及示范图片（见表 3-3）相符合的瑜伽动作图像标注为标准动作图片，而出现典型错误，即抬起脚放置在膝盖及以下的位置、抬起脚的脚趾不朝下、脊柱弯曲及上身躯干歪斜这几种情况的图片，标注为不标准动作图片。各类型典型错误及示例图片对照如表 3-5 所示。

表 3-5 树式各类型典型错误及示例图片

错误类型	抬起脚放置在膝盖及以下的位置	抬起脚的脚趾不朝下	脊柱弯曲	上身躯干歪斜
示例图片				

3.2.3 战士第二式

根据《瑜伽之光》中的体式讲解，艾扬格认为战士第二式应当分开两腿 120 到 135 厘米，两臂侧平举与肩齐，向两侧尽量延伸，手掌朝下，右脚右转 90 度，左脚也稍向右转，左腿伸直膝盖绷直，右大腿与地面平行，右胫骨与地板垂直，使右大腿与右小腿成直角，且右膝盖与右脚跟的连线垂直于地面，脸转向右侧，眼睛注视右掌，完全拉伸左腿后部的肌肉，使腿后部、脊背及臀部在一条直线上。网站 Everyday Yoga 中一篇讲解练习战士第二式的方法的文章^[60]认为练习时应当确保前膝盖和前脚踝

对齐，不要让膝盖向左漂移，因为这样会使膝关节拉伤。网站 Tint Yoga 中一篇讲解战士第二式练习注意要点的文章^[61]认为练习战士第二式时应当注意后肩不塌下来，两肩连线尽量平行于地面，并且要确保脊柱挺直不要太前倾，保持上身躯干垂直。

在战士第二式图片的图像标注过程中，将所有姿势关键点与《瑜伽之光》中提及的要点及示范图片（见表 3-3）相符合的瑜伽动作图像标注为标准动作图片，而出现典型错误，即后腿未伸直、前腿大腿不与地面平行、前腿小腿不与地面垂直及上身躯干没有挺直这几种情况的图片，标注为不标准动作图片。各类型典型错误及示例图片对照如表 3-6 所示。

表 3-6 战士第二式各类型典型错误及示例图片

错误类型	后腿未伸直	前腿大腿不与地面平行	前腿小腿不与地面垂直	上身躯干没有挺直
示例图片				

3.2.4 低位眼镜蛇式

根据《瑜伽之光》中的体式讲解，艾扬格认为低位眼镜蛇式应当伸直双腿，双脚相靠，膝盖绷直，脚趾向后，手掌放在骨盆区域附近，用手使劲按压地面，抬起躯干。网站 Everyday Yoga 中一篇讲解练习眼镜蛇式的方法的文章^[62]认为练习时应当将脚尖应该放在垫子上，不要把脚趾收起来，防止挤压脊椎。网站 Very Well Fit 中一篇关于如何练习眼镜蛇式的文章^[63]认为练习时肘部应该向后，而不是向两侧伸出，确保动作练习依靠的是背部肌肉，而不是手臂肌肉。根据对《瑜伽之光》中低位眼镜蛇示范动作的观察，可以看出抬起躯干的标准动作应当是把上身躯干抬起至接近挺直状态，才能很好地拉伸到背部肌肉，且抬起躯干时接触地面的位置约在盆骨附近。关于躯干抬起的触地部位，虽然《瑜伽之光》中在低位眼镜蛇的体式讲解没有强调，但在高位眼镜蛇的体式讲解中进行了强调，可认为是标准动作的要点之一。

在低位眼镜蛇式图片的图像标注过程中，将所有姿势关键点与《瑜伽之光》中提及的要点及示范图片（见表 3-3）相符合的瑜伽动作图像标注为标准动作图片，而出现典型错误，即手的摆放位置太前超过肩膀、躯干抬起不足、躯干抬起的触地部位不在盆骨附近以及膝盖未绷直这几种情况的图片，标注为不标准动作图片。各类型典型错误及示例图片对照如表 3-7 所示。

表 3-7 低位眼镜蛇式各类型典型错误及示例图片

错误类型	手的摆放位置太前超 过肩膀	躯干抬起不足	躯干抬起的触地部位不在盆 骨附近	膝盖未绷直
示例图片				

3.2.5 女神式

女神式这一瑜伽体式未收录入本文的瑜伽动作标准主要参考的权威书籍《瑜伽之光》中，因此动作标准的确认方法为综合参考一些瑜伽、运动内容相关文章中提及的女神式动作指导要点，通过总结共同点，针对矛盾点综合分析，确认数据集标注的动作要点标准。在确认女神式的动作标准过程中，分别参考了网站 Yoga Basics 中的文章《女神蹲》^[64]，网站 Beach Body On Demand 中的文章《如何做瑜伽中的女神式动作》^[65]，网站 Gaia 中的文章《女神式：Utkata Konasana》^[66]，网站 Be Extra Yoga 中的文章《如何做瑜伽女神式动作(Utkata Konasana)》^[67]，网站 Alomoves 中的文章《瑜伽类别：女神式》^[68]，以及网站 Everyday Yoga 中的文章《如何做瑜伽中的女神蹲》^[69]。

各文章中相一致的观点有：第一，双脚向外张开的角度。Yoga Basics，Beach Body On Demand 及 Gaia 中的三篇文章指出应呈 45 度各向斜前方，Be Extra Yoga，Alomoves 及 Everyday Yoga 中的三篇文章指出应朝向瑜伽垫的边角，由于双脚充分分开，可认为该角度也约为呈 45 度各向斜前方。第二，下蹲达到的深度。在提及这一要点的五篇文章中，Beach Body On Demand 中的文章指出应膝盖弯曲直到大腿与地面平行，Gaia 中的文章指出下蹲时应降低臀部努力接近膝盖的高度，Be Extra Yoga，Alomoves 及 Everyday Yoga 中的三篇文章指出下蹲时应努力使大腿与地面平行。综合来看，标准的女神式动作下蹲深度应当是大腿与地面平行或接近于平行状态的。第三，膝盖弯曲的方向。在提及这一要点的三篇文章，即 Beach Body On Demand，Gaia 和 Everyday Yoga 中的三篇文章都指出膝盖应该和脚趾对齐、方向相同。第四，躯干的状态。在提及这一要点的五篇文章中，Beach Body On Demand 中的文章指出将肩胛骨向后并向拉伸、抬起胸部，Gaia 中的文章指出应将尾骨拉向地板、保持脊柱伸长，Be Extra Yoga 中的文章指出应保持伸长脖子和脊椎，Alomoves 中的文章指出应挺直脊柱，Everyday Yoga 中的文章指出整个体式中应保持脊柱垂直、拉长躯干前部和腰部两侧。综合来看，标准女神式动作中的躯干状态应当是脊柱挺直、伸长、垂直于地面的状态。

各文章中有所不一致的观点有：第一，双脚分开的宽度。在提及这一要点的三篇文章中，Yoga

Basics 中的文章指出双脚应分开 3 英尺，Alomoves 中的文章指出应脚跟分开与肩同宽，Everyday Yoga 中的文章指出两脚应分开大约 4 英尺。每篇文章在该要点上的观点各不相同，然而考虑到个体差异，双脚分开的宽度确实不易有绝对的标准，而在达到双腿动作中的其他关键要点时，双脚分开的宽度显然是在合理状态的，因此在标注过程中可忽略这一要点的绝对要求，只对双腿动作的其他要点做好限制。第二，下蹲膝盖向两侧延伸出的长度。在提及这一要点的三篇文章中，Yoga Basics 中的文章指出弯曲膝盖应超过脚趾（然而在该文章给出的图片示范中膝盖并未超过脚趾，仅超过了脚踝，如图 3-1 所示），Beach Body On Demand 中的文章指出应使膝盖超过脚踝，Everyday Yoga 中的文章指出膝盖应弯曲到脚趾上方。由于 Yoga Basics 中的观点和示例图片在这一要点上并不一致，该观点可参考性值得怀疑，而另两篇文章中的观点，即膝盖应超过脚踝和膝盖在脚趾上方，这两个位置是很接近的，当膝盖超过脚踝时，就已接近脚趾上方的位置。因此，该要点将这些文章的观点综合总结为膝盖应超过脚踝。第三，手臂、手掌做出的姿态动作。Yoga Basics 和 Gaia 中的两篇文章所指出的手部动作都是弯曲肘部与肩齐高、手掌相对、指尖指向天空，Beach Body On Demand 中的文章指出的手部动作是双臂伸直、手掌朝内、做一个扫臂动作，而 Be Extra Yoga，Alomoves 及 Everyday Yoga 中的三篇文章都指出手部的动作有很多种类别，可以灵活多变。在女神式这一体式的练习中，手臂和手掌的动作对重要的肌肉力量使用影响微乎其微，再综合这 6 篇文章的观点，在标注过程中不将手臂、手掌动作视作区分动作标准与否的要点，不做统一标准，可由练习者自行决定。



图 3-1 Yoga Basics 中的文章《女神蹲》给出的示范图片^[63]

综上所述，可总结出图像标注时的参考女神式标准动作要点及标准动作图片示例如表 3-3 中所示。在女神式图片的图像标注过程中，将所有姿势关键点与整合总结的要点相符合的瑜伽动作图像标注为标准动作图片。而出现典型错误，即双脚向外张开的角度太小（接近于 0 度）、双脚向外张开的角度太大（接近于 90 度）、下蹲程度太小（离大腿与地面平行的程度还有明显差距）、膝盖未超过脚踝及脊柱未挺直这几种情况的图片，标注为不标准动作图片。各类型典型错误及示例图片对照如表 3-8 所示。

表 3-8 女神式各类型典型错误及示例图片

错误 类型	双脚向外张开的 角度太小	双脚向外张开的角度 太大	下蹲程度太小	膝盖未超过脚踝	脊柱未挺直
示例 图片					

4 基于多模态的瑜伽动作分析

为了推进瑜伽动作分析智能化，为瑜伽训练者提供智能瑜伽动作识别和动作评估，本章节展开瑜伽动作分析研究。本章瑜伽动作分析研究由基于多模态的瑜伽动作识别和基于多模态的瑜伽动作评估两项内容分别展开建立模型、展开实验。模型输入两个模态的数据，其一为通过 YogaCE 数据集中瑜伽动作图像的 RGB 数据，其二为利用 MediaPipe BlazePose 网络对 YogaCE 数据集中的瑜伽动作图片提取得到的人体骨架关键点坐标数据。多模态深度学习可以融合来自不同模态的信息从而提高模型的准确性，通过融合多个输入源的信息来平衡数据偏差增强模型的稳定性，本章采取这一方法，以获得瑜伽动作分析更优的模型。

4.1 模型设计

4.1.1 多模态网络设计

目前，通过将多个模态的特征在识别网络中融合已经成为了人体姿态动作检测中常用的方法之一，相较于单模态数据构建的模型，多模态之间的数据互补有助于模型表现出更好的精度。瑜伽动作识别模型 YogaRegNet 及瑜伽动作评估模型 YogaEvaNet 的输入数据即采用到两种模态，分别为瑜伽动作图片的 RGB 数据及瑜伽动作骨架关键点坐标数据。

多模态数据融合方法可划分为三种：基于数据的多模态数据融合、基于特征的多模态数据融合和基于决策的多模态数据融合^[70]，这三种融合方法各有其优势及劣势。为了获得最佳的瑜伽动作识别模型，本实验针对以上三种多模态融合方法分别设计了相应的瑜伽动作多模态融合方案，具体融合方法如下。

4.1.1.1 数据层融合

在数据层融合方案中，将 MediaPipe BlazePose 网络对 YogaCE 数据集中的瑜伽动作图片提取得到的人体骨架数据通过以人体关键点坐标为点、以人体关键点之间的有效连线为线，通过 MediaPipe 自带的可视化功能在图像数据上绘制，获得提取得到的人体骨架在原图像上进行了添加、绘制的人体骨架可视化瑜伽动作图片。人体骨架可视化瑜伽动作图片示例图 4-1 所示。



图 4-1 人体骨架可视化瑜伽动作图片示例

数据层融合方案就以这份原图像数据集转化得到的人体骨架可视化瑜伽动作图片数据集作为输入数据，在 VGG 网络上进行训练，从而得到该实验方案下的分类模型。该 VGG 网络结构见 4.1.2.2。该过程如图 4-2 所示。

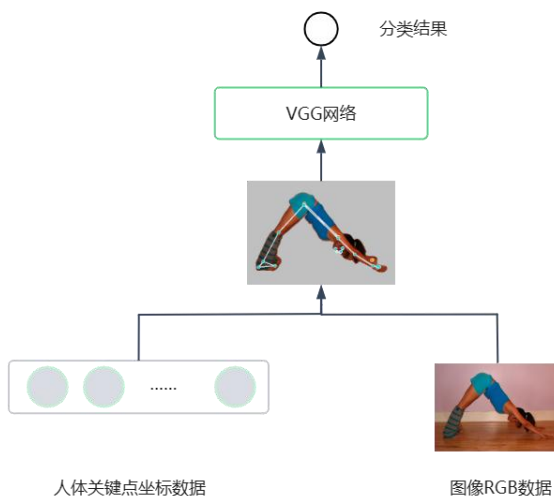


图 4-2 数据层融合方案模型结构图

4.1.1.2 特征层融合

在特征层融合方案中，设计网络分为 MLP 网络分支及 VGG 网络分支，两分支得到的结果拼接，输入输出结果的全连接层中。将 MediaPipe BlazePose 网络对 YogaCE 数据集中的瑜伽动作图片进行提取得到的人体关键点坐标数据输入到 MLP 网络分支中，将 YogaCE 数据集的 RGB 数据输入到 VGG 网络分支中，两分支在全连接层进行拼接，并输出结果，从而整合为一个分类模型。该 MLP 及 VGG 融合网络结构见 4.1.2.3。该过程如图 4-3 所示。

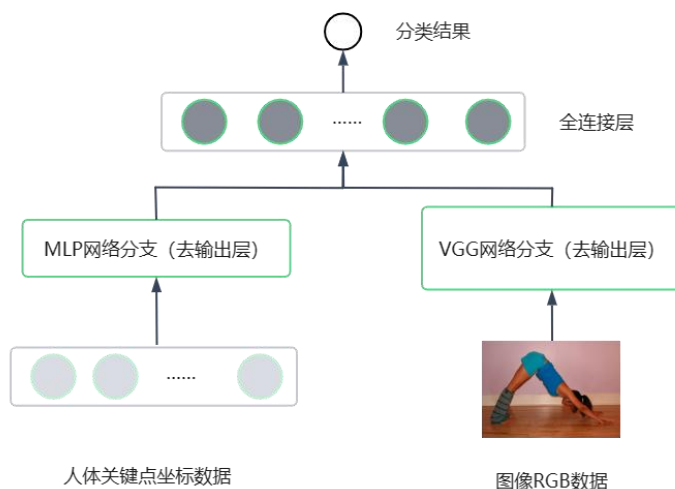


图 4-3 特征层融合方案模型结构图

4.1.1.3 决策阶段融合

在决策阶段融合方案中，将 MediaPipe BlazePose 网络对 YogaCE 数据集中的瑜伽动作图片进行提取得到的人体关键点坐标数据输入到 MLP 网络中，将 YogaCE 数据集的 RGB 数据输入到 VGG 网络中，作为两个分类模型。该 VGG 网络结构见 4.1.2.2，该 MLP 网络结构见 4.1.2.1。将两分类模型输出的分类结果按照一定权重进行决策整合，得到最终的分类结果。该权重的选取在实验过程中选择不同权重得到的精度对比，选取值确定为精度最佳的权重值。该过程如图 4-4 所示。

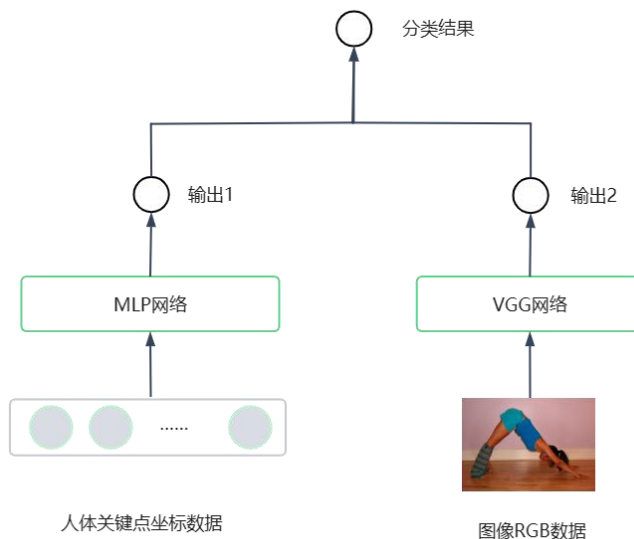


图 4-4 决策阶段融合方案模型结构图

4.1.2 主干网络设计

在以上多模态模型设计中，涉及到 MLP 网络、VGG 网络、MLP 分支及 VGG 分支融合网络这 3 种主干网络，本小节对这 3 种主干网络分别展开详细介绍。

4.1.2.1 三层 MLP 网络

决策阶段融合模型（见 4.1.1.3）中采用的网络之一是一个三层的 MLP 网络，该网络由一个输入层，一个隐藏层，一个输出层构成。输入层根据输入骨架坐标数据的维度由 99 个神经元组成，隐藏层由 69 个神经元组成，在瑜伽动作识别模型这一五分类模型中输出层由 5 个神经元组成，在瑜伽动作评估模型这一二分类模型中输出层则由 2 个神经元组成。瑜伽动作识别模型的 MLP 网络结构及各层神经元数量如图 4-5 所示，瑜伽动作评估模型的 MLP 网络结构与之类似，仅将输出层神经元个数替为 2 个。

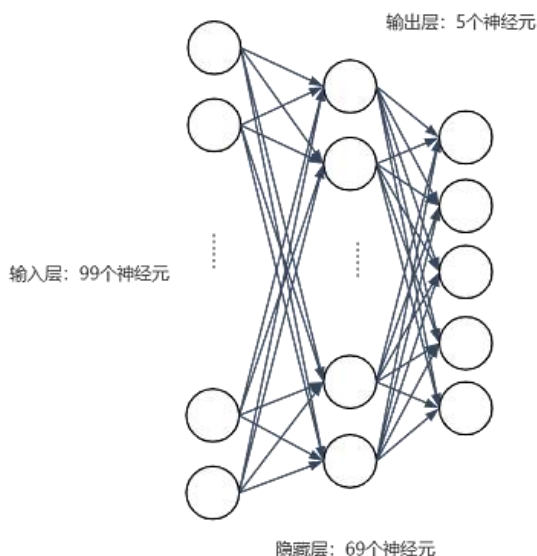


图 4-5 瑜伽动作识别模型三层 MLP 网络结构及各层神经元数量

其中，网络在隐藏层后设置了节点随机抛弃，以及 Relu 激活函数进行激活，来防止过拟合。

4.1.2.2 VGG19 网络

数据层融合模型（见 4.1.1.1）中采用的网络及决策阶段融合模型（见 4.1.1.3）中采用的网络之一是经典图像识别网络 VGG19 网络，该网络通过反复的使用 3x3 的小型卷积核和 2x2 的最大池化层构筑了一个 19 层深的卷积神经网络。瑜伽动作评估模型的 VGG19 网络具体网络结构如图 4-6 所示，瑜伽动作评估模型的 VGG19 网络结构与之类似，仅将输出层神经元个数替为 2 个。

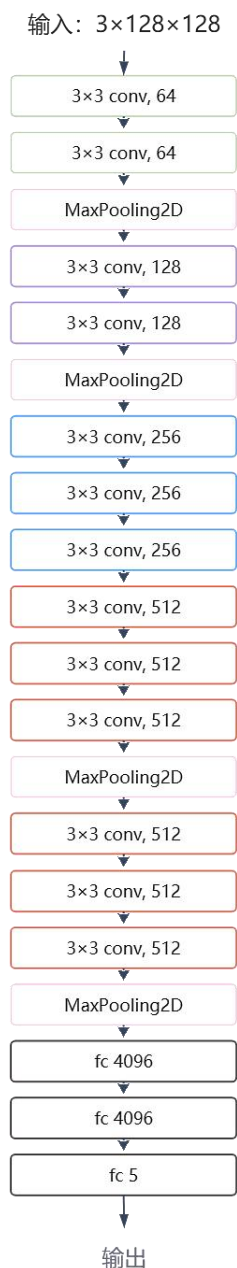


图 4-6 瑜伽动作识别模型的 VGG19 网络结构

其中，每个卷积层后使用 Relu 激活函数进行激活。前两个全连接层后分别设置了 drop out 函数随机丢弃，以及 Relu 激活函数进行激活。在实验中，VGG19 网络使用到了 torchvision^[71]给出的 VGG19 在 ImageNet-1K 这一规模庞大的图像分类数据集上进行了预训练的预训练模型，来增强识别效果，提升精度，对于这一模型仅在最后输出层的输出神经元个数微调为 5 或 2。

4.1.2.3 MLP 分支及 VGG 分支融合网络

特征层融合（见 4.1.1.2）中采用的网络由上述三层 MLP 网络作为 MLP 分支，及上述 VGG19 网络作为 VGG 分支，在最后的全连接层拼接而成，即将 RGB 数据输入 VGG19 网络，将骨架坐标数据输入 MLP 网络，取这两个网络各自最后一个全连接层（即输出层）之前的输入数据进行拼接，将拼接后得到的数据输入两分支合并后共同的输出层。瑜伽动作评估模型的融合网络具体网络结构如图 4-7 所示，瑜伽动作评估模型的 VGG19 网络结构与之类似，仅将输出层神经元个数替为 5 或 2。



图 4-7 瑜伽动作识别模型的 MLP 及 VGG 融合网络结构

其中，MLP 分支在隐藏层后设置了节点随机抛弃，以及 Relu 激活函数进行激活，且在该网络中使用到已经用骨架坐标数据训练集在 4.1.2.1 中的三层 MLP 网络上训练了 7000 个 epoch 的预训练模型，仅删去了最后一个全连接层，并在该分支后设置一个全连接层使之与 VGG 分支合并。VGG 分支在每个卷积层后使用 Relu 激活函数进行激活。前两个全连接层后分别设置了 drop out 函数随机丢弃，以及 Relu 激活函数进行激活。在实验中，VGG19 网络使用到了 torchvision 给出的 VGG19 在 ImageNet-1K 这一规模庞大的图像分类数据集上进行了预训练的预训练模型，仅删去了最后一个全连接层，并在该分支后设置一个全连接层使之与 VGG 分支合并。

4.2 实验结果与分析

4.2.1 数据集准备

4.2.1.1 数据集组成

瑜伽动作识别模型 YogaRegNet 的训练和测试用到的图像数据集为自整理标注数据集 YogaCE（详见第 3 节的数据集介绍）合并同一体式中的标准和 not standard 动作图片，根据瑜伽动作类别的不同共分为 5 类。由于在使用 MediaPipe BlazePose 提取人体关键点时有一张低位眼镜蛇式动作图片未检测到人像，实验使用到的数据量比原数据集 YogaCE 少一张图片。训练集及测试集按 7: 3 的比例划分，最终各类别瑜伽动作的训练集、测试集及总图片数量如表 4-1 所示。

表 4-1 各类别瑜伽动作的训练集、测试集及总图片数量

图片类型	下犬式	树式	战士二式	低位眼镜蛇式	女神式
训练集图片数量	164	143	137	166	135
测试集图片数量	72	63	60	73	59
图片总数	236	206	197	239	194

瑜伽动作评估模型 YogaEvaNet 的训练和测试用到的图像数据集 YogaCE，根据其中的 5 种动作类别划分出 5 个数据集，每个数据集分为标准动作和 not standard 动作两个类别。由于在使用 MediaPipe BlazePose 提取人体关键点时有一张低位眼镜蛇式动作图片未检测到人像，实验使用到的数据量比原数据集 YogaCE 少一张图片。训练集及测试集按 7: 3 的比例划分，最终各类别瑜伽动作数据集的标准动作训练集图片、not standard 动作训练集图片、标准动作测试集图片、not standard 动作测试集图片及总图片数量如表 4-2 所示。

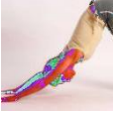
表 4-2 各类别瑜伽动作数据集的标准/不标准动作训练集/测试集图片及总图片数量

图片类型	下犬式	树式	战士二式	低位眼镜蛇式	女神式
标准动作训练集图片数量	91	86	58	88	64
不标准动作训练集图片数量	73	57	79	78	71
标准动作测试集图片数量	40	38	25	39	28
不标准动作测试集图片数量	32	25	35	34	31
图片总数	236	206	197	239	194

4.2.1.2 图像模态数据预处理

对原始数据进行图像增广预处理可以扩大训练集的规模，并且降低模型对某些属性的依赖，从而提高模型的泛化能力。对于 4.2.1.1 中的训练集，在转化为 tensor 时除了使用到仅经过大小改变为 128*128 和归一化过程的数据，还扩充了额外进行对比度在 0.5-2 之间随机取值变化、在 20° 内随机放射变化、亮度在 0.5-2 内随机取值进行变化、饱和度在 0.5-2 内随机取值进行变化、随机裁剪（在 300*300 的图片中随机裁剪 128*128）、概率直方图均衡以及水平翻转这七种图像处理转化的数据。仅经过大小改变为 128*128 和归一化过程的原图像及还经过以上图像变化的图像示例如表 4-3 所示。

表 4-3 原图像及还经过其他图像变化的图像示例

图像变化 情况	原图像	对比度在 0.5-2 之 间随机取 值变化	在 20° 内 随机放射 变化	亮度在 0.5-2 内 随机取值 进行变化	饱和度在 0.5-2 内 随机取值 进行变化	随机裁剪	概率直方 图均衡	水平翻转
示例								

4.2.1.3 骨架模态数据获取

人体骨架数据即通过 MediaPipe BlazePose 技术（详见 2.2）提取图片中的体关键点三维坐标，共得到 33 个关键点（详见 2.2.1.2）的三维坐标，即每张图片转化为 99 个维度的数据。该过程通过 MdiaPipe 官网^[46]提供的 python 函数检测图片中的人体关键点坐标并保存至本地实现。人体关键点坐标数据的测试集、训练集及总数数据量同 4.2.1.1 中的图像数据。

在获取 4.2.1.1 中的图像数据的人体关键点坐标并保存的同时，使用 MdiaPipe 官网提供的 python 函数在原图像数据上绘制了人体坐标关键点及其连线（绘制所得的图片效果见 2.2.2），获得提取得到的人体骨架在原图像上进行了添加、绘制的人体骨架可视化瑜伽动作图片数据集。体骨架可视化瑜伽动作图片数据集的测试集、训练集及总数数据量同 4.2.1.1 中的图像数据。

4.2.2 实验设置

4.2.2.1 损失函数

本章所有模型实验使用到的损失函数均为交叉熵损失函数^[45]（Cross-Entropy Loss Function）。交叉熵损失函数常用于分类问题中，通过比较模型预测结果和真实结果之间的差距，来评估模型的性能好坏。交叉熵损失函数的本质是对真实结果和预测结果之间的差距进行量化，并将其转化为一个数值。在使用交叉熵损失函数进行训练时，我们的目标是通过调整模型参数，使得损失函数的值尽可能小。通过随机梯度下降等优化算法，可以对模型参数进行更新，以降低损失函数的值，从而提高模型的预测准确率。

交叉熵损失函数在多分类问题中的公式可表示为：

$$L(y, \hat{y}) = - \sum_{i=1}^n y_i \log(\hat{y}_i) \quad (4-1)$$

其中， n 表示类别数量， y_i 表示第 i 个类别的实际标签， $\hat{y}_i$ 表示第 i 个类别的实际预测概率值。

4.2.2.2 其他实验设置

为了方便更好地监测训练过程，及避免训练过程中可能出现中断的意外，实验对模型多次分散的训练，每次训练在上一次训练得到的模型基础上继续进行训练。每个实验都在损失函数收敛后终止实验，因此每个实验最终训练的 epoch 数即实验损失函数收敛时的已训练的 epoch 数量。以下内容介绍瑜伽动作识别模型即瑜伽动作评估模型在训练过程中重要的一些参数设置。

4.2.2.2.1 瑜伽动作识别模型其他实验设置

瑜伽动作识别模型实验分为 3 个双模态实验（数据层融合实验、特征层融合实验及决策阶段融合实验）和 2 个单模态对照实验（RGB 图像数据输入 VGG 网络实验及人体骨架关键点坐标数据输入 MLP 网络实验）。其中数据层融合实验、特征层融合实验、RGB 图像数据输入 VGG 网络实验及人体骨架关键点坐标数据输入 MLP 网络实验这 4 个实验种各训练了一个网络，这些实验在模型训练时的学习率设置、全部训练的 epoch 数量、分为多少次训练及每次训练的 epoch 数量如表 4-4 所示。

表 4-4 瑜伽动作识别模型实验数据设置

设置参数	数据层融合	特征层融合	RGB 图像数据输入 VGG 网络	人体骨架关键点坐 标数据输入 MLP 网 络
学习率	10^{-5}	4×10^{-7}	2×10^{-5}	4×10^{-7}
全部 epoch 数量	20	200	70	7000
训练分散次数	1	3	1	1
每次训练的 epoch 数量	20	40/80/80	70	7000

4.2.2.2.2 瑜伽动作评估模型其他实验设置

瑜伽动作评估实验针对五个动作类别分别训练模型，每个类别展开 5 组实验，分为 3 个双模态实验（数据层融合实验、特征层融合实验及决策阶段融合实验）和 2 个单模态对照实验（RGB 图像数据输入 VGG 网络实验及人体骨架关键点坐标数据输入 MLP 网络实验），其中数据层融合实验、特征层融合实验、RGB 图像数据输入 VGG 网络实验及人体骨架关键点坐标数据输入 MLP 网络实验这 4 个实验种各训练了一个网络，以上所述 5 组实验各组的 4 个实验所涉及参数设置，即在模型训练时的学习率设置、全部训练的 epoch 数量、分为多少次训练及每次训练的 epoch 数量如表 4-5、表 4-6、表 4-7、表 4-8、表 4-9 所示。

表 4-5 下犬式瑜伽动作评估模型实验数据设置

设置参数	数据层融合	特征层融合	RGB 图像数据输入 VGG 网络	人体骨架关键点坐 标数据输入 MLP 网 络
学习率	4×10^{-7}	4×10^{-7}	4×10^{-7}	4×10^{-7}
全部 epoch 数量	250	150	150	2000
训练分散次数	5	3	3	1
每次训练的 epoch 数量	50	50	50	2000

表 4-6 女神式作评估模型实验数据设置

设置参数	数据层融合	特征层融合	RGB 图像数据输入 VGG 网络	人体骨架关键点坐 标数据输入 MLP 网 络
学习率	4×10^{-7}	4×10^{-7}	4×10^{-7}	4×10^{-7}
全部 epoch 数量	250	150	150	2000
训练分散次数	5	3	3	1
每次训练的 epoch 数量	50	50	50	2000

表 4-7 地位眼镜蛇动作评估模型实验数据设置

设置参数	数据层融合	特征层融合	RGB 图像数据输入 VGG 网络	人体骨架关键点坐 标数据输入 MLP 网 络
学习率	4×10^{-7}	4×10^{-7}	4×10^{-7}	4×10^{-7}
全部 epoch 数量	150	150	150	2000
训练分散次数	3	3	3	1
每次训练的 epoch 数量	50	50	50	2000

表 4-8 树式瑜伽动作评估模型实验数据设置

设置参数	数据层融合	特征层融合	RGB 图像数据输入 VGG 网络	人体骨架关键点坐 标数据输入 MLP 网 络
学习率	4×10^{-7}	4×10^{-7}	4×10^{-7}	4×10^{-7}
全部 epoch 数量	150	150	150	2000
训练分散次数	3	3	3	1
每次训练的 epoch 数量	50	50	50	2000

表 4-9 战士第二式瑜伽动作评估模型实验数据设置

设置参数	数据层融合	特征层融合	RGB 图像数据输入 VGG 网络	人体骨架关键点坐 标数据输入 MLP 网 络
学习率	4×10^{-7}	4×10^{-7}	4×10^{-7}	4×10^{-7}
全部 epoch 数量	150	200	150	2000
训练分散次数	3	4	3	1
每次训练的 epoch 数量	50	50	50	2000

4.2.3 评价标准

网络分类识别的性能主要使用总体准确率及各类别各自的准确率来综合评价。同时结合模型参数量，判断模型所需的计算资源，辅助判断模型的适用情况。

4.2.3.1 瑜伽动作识别准确率

假定瑜伽动作识别这一 5 分类问题的混淆矩阵及各实际类别和预测类别对应数量如表 4-10 所示。

表 4-10 瑜伽动作类别分类的混淆矩阵及各实际类别和预测类别对应数量

预测类别/实际类别	下犬式	女神式	地位眼镜蛇式	树式	战士二式
下犬式	TP ₀	FP _{0,0}	FP _{0,1}	FP _{0,2}	FP _{0,3}
女神式	FP _{1,0}	TP ₁	FP _{1,1}	FP _{1,2}	FP _{1,3}
地位眼镜蛇式	FP _{2,0}	FP _{2,1}	TP ₂	FP _{2,2}	FP _{2,3}
树式	FP _{3,0}	FP _{3,1}	FP _{3,2}	TP ₃	FP _{3,3}
战士二式	FP _{4,0}	FP _{4,1}	FP _{4,2}	FP _{4,3}	TP ₄

则模型识别的总体准确率 Accuracy 的计算公式为：

$$\text{Accuracy} = \frac{\sum_{i=0}^4 \text{TP}_i}{\sum_{i=0}^4 \text{TP}_i + \sum_{i=0}^4 \sum_{j=0}^3 \text{FP}_{i,j}} \quad (4-2)$$

各类别的准确率 Accuracy_i 的计算公式为：

$$\text{Accuracy}_i = \frac{\text{TP}_i}{\text{TP}_i + \sum_{j=0}^3 \text{FP}_{i,j}}, i=0,1,2,3,4 \quad (4-3)$$

其中，Accuracy₁，Accuracy₂，.....，Accuracy₄分别指类别下犬式、女神式、地位眼镜蛇式、树式、战士二式在模型中的识别准确率。

4.2.3.2 瑜伽动作评估准确率

假定瑜伽动作评估这一 2 分类问题的“标准”这一分类表示预测为正样本（positive），“不标准”这一分类表示预测为负样本（negative），则该模型预测类别的混淆矩阵及各实际类别和预测类别对应数量如表 4-11 所示。

表 4-11 瑜伽动作标准性评估的混淆矩阵及各实际类别和预测类别对应数量

预测类别/实际类别	标准	不标准
标准	TP	FN
不标准	FP	TN

则模型识别的总体准确率 Accuracy 的计算公式为：

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (4-4)$$

识别标准动作图片的准确率为：

$$\text{Accuracy}_P = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4-5)$$

识别不标准动作图片的准确率为：

$$\text{Accuracy}_N = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (4-6)$$

4.2.3.3 模型参数量

参数量是指模型的所有带参数的层的权重参数总量，体现了算法的时间复杂度。

4.1.2 中各网络包含了卷积层、池化层、全连接层，其中，池化层没有参数，即：

$$\text{Params}_{\text{pooling}} = 0 \quad (4-7)$$

池化层的参数量计算公式为：

$$\text{Params}_{\text{conv}} = K_2 \times C_i \times C_o + C_o \quad (4-8)$$

其中，K 为积大小， C_i 为输入通道数， C_o 为输出通道数。

全连接层的计算公式为：

$$\text{Params}_{fc} = T_i \times T_o + T_o \quad (4-9)$$

其中， T_i 为输入向量长度， T_o 为输出向量长度。

总参数量 Params 为所有层参数量的总和，即在 4.1.2 中各网络的总参数量可表示为：

$$\text{Params} = \sum \text{Params}_{\text{pooling}} + \sum \text{Params}_{\text{conv}} + \sum \text{Params}_{fc} \quad (4-10)$$

4.2.4 实验结果及分析

4.2.4.1 瑜伽动作识别

4.2.4.1.1 实验结果

实验构建的对照实验除了根据不同的多模态融合方案构建的三种模型，还有两个对照组实验，分别为仅输入图像数据的 VGG 网络以及仅输入骨架坐标数的 MLP 网络。根据以上五组实验进行对比，可以确认多模态融合方法相对于单模态方法的有效性，并从三种多模态实验方案中选取最优的一种作为最终的分类模型。

上述五组实验的总准确率及各分类类别各自的准确率如表 4-12 所示。

表 4-12 瑜伽动作识别五组实验的总准确率及各分类类别各自的准确率

融合策略	总准确率	下犬式准确率	女神式准确率	地位眼镜蛇式准确率	树式准确率	战士第二式准确率
数据层融合	99%	100%	98%	100%	98%	100%
特征层融合	95%	100%	88%	98%	95%	93%
决策阶段融合	98%	100%	98%	100%	98%	93%
仅输入图像数据的 VGG 网络	97%	100%	100%	97%	96%	93%
仅输入骨架坐标数据的 MLP 网络	86%	95%	74%	90%	88%	78%

同上中所述的五组实验，其各自的模型参数量如表 4-13 所示。

表 4-13 瑜伽动作识别五组实验的模型参数量

融合策略	数据层融合	特征层融合	决策阶段融合	仅输入图像数据的 VGG 网络	仅输入骨架坐标数据的 MLP 网络
参数量	139590725	139597970	139597975	139590725	7250

4.2.4.1.2 结果分析

与由单模态数据构建的两种模型，仅输入图像数据的 VGG 网络瑜伽动作识别模型，及仅输入骨架坐标数据的 MLP 网络瑜伽动作识别模型相比，三种双模态数据构建的瑜伽动作识别模型中，数据层融合模型和决策阶段融合模型的精度性能都超过了单模态数据模型，特征层融合模型的精度虽然略低于仅输入图像数据的 VGG 网络模型，但显著高于仅输入骨架坐标数据的 MLP 网络。综上所述，多模态融合的方法在瑜伽动作识别模型的构建中展现出了其优势。

在三种双模态数据构建的瑜伽动作识别模型的参数量接近的情况下，数据层融合模型的精度最优，决策阶段融合模型次之，特征层融合模型最劣。可能的原因分析如下。

数据层融合模型的优势在于将瑜伽动作图像的两个模态在数据层进行了融合，直接对原始数据进行处理，因此信息损失量小，能够使模型注意到最微小的信息，因此精度最高。特征层融合模型的数据融合发生在全连接层，此时的数据已发生了一些信息损失，此时的数据融合可能导致无法有效地提升模型精度水平，尤其是在女神式和战士第二式这两个仅输入骨架坐标数据的 MLP 网络模型的弱势识别分类展现了其劣势。决策阶段融合模型将两个单模态数据构建的模型的输出融合进行识别决策，虽然也达到了与数据层融合模型接近的精度，但依然没有达到数据层融合模型信息损失量足够小的优势，尤其是战士第二式的识别性能显著受到了仅输入骨架坐标数据的 MLP 网络模型的影响，精度不佳，因此精度表现次之。

同时，在整体精度最佳的数据层融合模型中，识别下犬式、地位眼镜蛇式、战士第二式的准确率最好，达 100%，识别女神式、树式的效果次之，为 98%。

三种多模态融合模型的参数量大小接近，因此不对最终模型方案的选取形成影响，而是仅需考虑模型的识别精度。

综上所述，瑜伽动作识别模型 YogaRegNet 选取数据层融合模型作为最终的模型方案，以达到最优的识别效果。

4.2.4.1.3 参数讨论

在决策阶段融合实验中，将图像数据输入经过图像数据训练的 VGG 网络得到一个输出 `outputs1`，将骨架坐标数据输入经过骨架坐标数据训练的 MLP 网络得到一个输出 `outputs2`，将这两个输出进行融合，使用到的方法为按一定权重将这两个输出结果相加得到用于进行分类决策的输出结果 `outputs`，其算术关系为：

$$\text{outputs} = \text{outputs1} * \text{we} + \text{outputs2} \quad (4-11)$$

其中，`we` 为两个输出结果相加时 `outputs1` 的权重，已知经过图像数据训练的 VGG 网络比经过骨架坐标数据训练的 MLP 网络精度更高，因此对于 `we`，在 1, 2, 3, ..., 10 之间进行取值，观察决策阶段融合模型测试结果中精度的变化来选取最佳的权重值。根据实验结果，模型精度与权重值 `we` 的关系如图 4-8 所示。

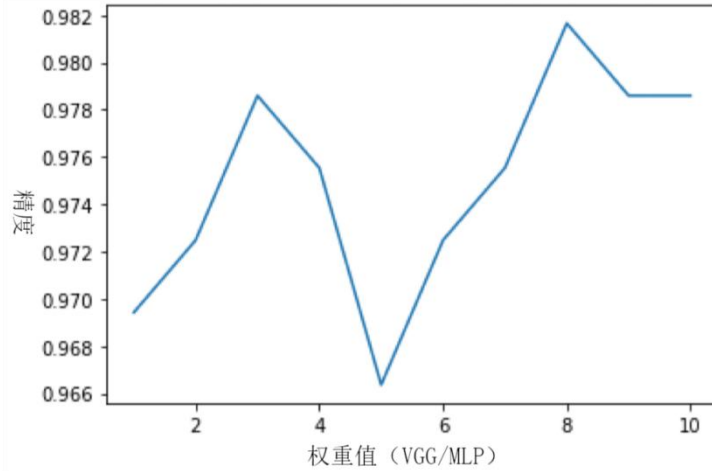


图 4-8 模型精度与权重值 w_e 的关系

根据这一结果对比，最终选取 8 作为 w_e 的取值。

4.2.4.2 瑜伽动作评估

4.2.4.2.1 实验结果

针对五组待给出评估分析模型的瑜伽体式，分别进行实验，每组实验构建的对照实验除了根据不同的多模态融合方案构建的三种模型及两个对照组实验。根据这五组实验进行对比，可以在每组体式中确认多模态融合方法相对于单模态方法的有效性，并从三种多模态实验方案中选取最优的一种作为最终的分类模型。

下犬式、女神式、低位眼镜蛇式、树式、战士第二式这 5 种体式的五个实验的总准确率及各分类类别各自的准确率如表 4-14、表 4-15、表 4-16、表 4-17、表 4-18 所示。

表 4-14 瑜伽动作评估下犬式 5 个实验的总准确率及各分类类别各自的准确率

融合策略	总准确率	不标准动作准确率	标准动作准确率
数据层融合	71%	55%	85%
特征层融合	70%	62%	77%
决策阶段融合	76%	65%	85%
仅输入图像数据的 VGG 网络	70%	56%	82%
仅输入骨架坐标数据的 MLP 网络	68%	65%	70%

表 4-15 瑜伽动作评估女神式 5 个实验的总准确率及各分类类别各自的准确率

融合策略	总准确率	不标准动作准确率	标准动作准确率
数据层融合	71%	77%	64%
特征层融合	72%	70%	75%
决策阶段融合	76%	70%	82%
仅输入图像数据的 VGG 网络	72%	67%	78%
仅输入骨架坐标数据的 MLP 网络	61%	58%	64%

表 4-16 瑜伽动作评估地位眼镜蛇式 5 个实验的总准确率及各分类类别各自的准确率

融合策略	总准确率	不标准动作准确率	标准动作准确率
数据层融合	76%	67%	84%
特征层融合	72%	70%	74%
决策阶段融合	80%	67%	92%
仅输入图像数据的 VGG 网络	69%	55%	82%
仅输入骨架坐标数据的 MLP 网络	73%	64%	82%

表 4-17 瑜伽动作评估树式 5 个实验的总准确率及各分类类别各自的准确率

融合策略	总准确率	不标准动作准确率	标准动作准确率
数据层融合	69%	68%	71%
特征层融合	66%	60%	71%
决策阶段融合	68%	56%	76%
仅输入图像数据的 VGG 网络	65%	56%	71%
仅输入骨架坐标数据的 MLP 网络	60%	36%	76%

表 4-18 瑜伽动作评估战士第二式 5 个实验的总准确率及各分类类别各自的准确率

融合策略	总准确率	不标准动作准确率	标准动作准确率
数据层融合	80%	74%	88%
特征层融合	73%	74%	72%
决策阶段融合	75%	68%	84%
仅输入图像数据的 VGG 网络	75%	65%	88%
仅输入骨架坐标数据的 MLP 网络	65%	68%	60%

五组体式的各实验模型参数量一致，其中的 5 个实验各自的模型参数量如表 4-19 所示。

表 4-19 瑜伽动作评估 5 个实验的模型参数量

融合策略	数据层融合	特征层融合	决策阶段融合	仅输入图像数据 的 VGG 网络	仅输入骨架坐标 数据的 MLP 网络
参数量	139578434	139585265	139585270	139578434	6836

4.2.4.2.2 结果分析

在 5 个体式的实验中，下犬式、女神式、树式的三组双模态数据构建的瑜伽动作评估模型总准确

率都超过了两组单模态数据构建的模型或与之持平。地位眼镜蛇式的特征层融合模型总准确率虽低于其仅输入骨架坐标数据的 MLP 网络模型，但高于其仅输入图像数据的 VGG 网络模型。战士第二式的特征层融合模型总准确率虽低于其仅输入图像数据的 VGG 网络模型，但高于其仅输入骨架坐标数据的 MLP 网络模型。综合来看，双模态数据融合在瑜伽动作评估的五种体式模型建构中展现了其优势。

每个体式的 5 组实验中，下犬式、女神式、低位眼镜蛇式都在决策阶段融合模型中获得了最高的总准确率，树式、战士第二式则在数据层融合模型中获得了最高的总准确率。与瑜伽动作识别模型类似，数据层融合的方法在树式和战士第二式的动作评估中因其信息损失量小的特点展现出了精度优势。而决策阶段融合根据最佳的权重对两个模态数据各自在其网络中的输出的整合，在二分类问题中能够更多地给出正确的分类结果，在下犬式、女神式、低位眼镜蛇式的动作评估中展现了其优势。

同时，在以上所用实验结果中，仅女神式的数据层融合模型和战士第二式的仅输入骨架坐标数据的 MLP 网络模型在评估不标准动作时的准确率高于评估标准动作时的准确率，其余实验模型均在评估标准动作时的准确率高于评估不标准动作时的准确率。每组体式实验的精度最佳模型都是在评估标准动作时的准确率高于评估不标准动作时的准确率。综合来看，瑜伽动作评估模型将标准瑜伽动作图片正确分类的能力高于将不标准瑜伽动作图片正确分类的能力。

三种多模态融合模型的参数量大小接近，因此不对最终模型方案的选取形成影响，而是仅需考虑模型的识别精度。

综上所述，瑜伽动作评估模型 YogaEvaNet 的下犬式、女神式、低位眼镜蛇式模型选取决策阶段融合模型作为最终的模型方案，树式、战士第二式模型选取数据层融合模型作为最终的模型方案，以达到最优的评估效果。

4.2.4.2.3 参数讨论

在决策阶段融合实验中，将图像数据输入经过图像数据训练的 VGG 网络得到一个输出 `outputs1`，将骨架坐标数据输入经过骨架坐标数据训练的 MLP 网络得到一个输出 `outputs2`，将这两个输出进行融合，使用到的方法为按一定权重将这两个输出结果相加得到用于进行分类决策的输出结果 `outputs`。对于下犬式、女神式、树式、战士第二式这 4 组实验的决策阶段融合实验，其算术关系为：

$$\text{outputs} = \text{outputs1} * \text{we} + \text{outputs2} \quad (4-12)$$

其中，`we` 为两个输出结果相加时 `outputs1` 的权重，已知经过图像数据训练的 VGG 网络比经过骨架坐标数训练的 MLP 网络精度更高，因此对于 `we`，在 1, 2, 3, ..., 10 之间进行取值，观察决策阶段融合模型测试结果中精度的变化来选取最佳的权重值。

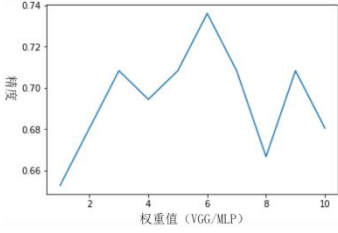
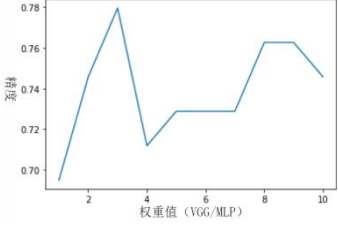
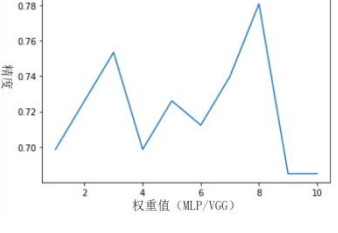
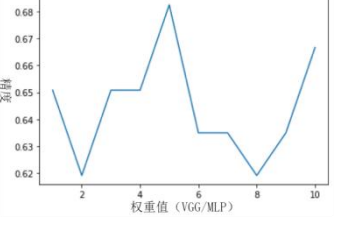
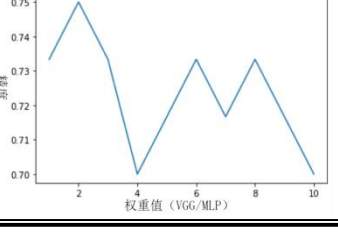
对于地位眼镜蛇式的决策阶段融合实验，其算术关系为：

$$\text{outputs} = \text{outputs1} + \text{outputs2} * \text{we} \quad (4-13)$$

其中，`we` 为两个输出结果相加时 `outputs2` 的权重，已知经过骨架坐标数训练的 MLP 网络比经过图像数据训练的 VGG 网络精度更高，因此对于 `we`，在 1, 2, 3, ..., 10 之间进行取值，观察决策阶段融合模型测试结果中精度的变化来选取最佳的权重值。

根据实验结果，每组体式实验的模型精度与权重值 w_e 的关系图及最终选取的 w_e 值如表 4-20 所示。

表 4-20 瑜伽动作评估每组体式决策阶段融合实验模型精度与权重值 w_e 的关系图及最终选取的 w_e 值

体式类别	模型精度与权重值 w_e 的关系图	最终选取的 w_e 值
下犬式		6
女神式		3
地位眼镜蛇式		8
树式		5
战士二式		2

4.3 本章小结

本章介绍了瑜伽动作分析实验中使用到的 3 种多模态融合方案，及其中设计到的 3 种主干网络结构。同时说明了数据集准备、实验设置及结果的评价标准。实验结果显示多模态融合模型在瑜伽动作

分析中相较于单模态数据模型表现出更好的精度。瑜伽动作识别模型 YogaRegNet 最终选取了整体精度最高的数据层融合实验模型，该模型在识别下犬式、低位眼镜蛇式、战士第二式的准确率最好，识别女神式、树式的效果次之，整体准确率为 99%。瑜伽动作评估模型 YogaEvaNet 的下犬式、女神式、低位眼镜蛇式模型选取了整体精度最高的决策阶段融合模型作为最终的模型方案，树式、战士第二式模型选取整体精度最高的数据层融合模型作为最终的模型方案。这些模型评估标准动作图片的标准性性能优于评估不标准动作图片标准性的性能。下犬式、女神式、低位眼镜蛇式、树式、战士第二式的最终选取模型整体准确率分别为 76%、76%、80%、69%、80%。

5 总结与展望

5.1 工作总结

本文对图像识别、人体姿态估计、瑜伽动作的识别与评估的研究背景与意义进行了总结与说明，阐述了目前的研究现状及存在的问题。介绍了本文模型算法涉及到的卷积神经网络、MLP 网络的基本结构及神经网络训练的重要过程，介绍了基于 MediaPipe BlazePose 的人体关节点提取这一技术的基本原理和其神经网络的主要结构。阐明了实验使用到的自整理标注数据集 YogaCE 的数据来源、选取标准及图像标注的判断依据。基于神经网络的理论基础，结合当前现有的网络和常见的多模态数据融合方法，提出了三种多模态网络，分别为数据层融合网络、特征层融合网络及决策阶段网络，通过实验选取精度最优的模型，实现了瑜伽动作识别网络 YogaRegNet 及 5 种体式的瑜伽动作评估网络 YogaEvaNet。研究工作与成果具体如下：

(1) 利用网络上现有的瑜伽动作分类数据集，参考瑜伽动作体式标准经典书籍及网络上的相关资料，整理标注了瑜伽动作分类评价数据集 YogaCE。相较于现有的仅标注了动作类别的瑜伽动作图片数据集，该数据集增加了瑜伽动作标准与否的分类标注，为瑜伽动作评估模型的研究提供了一份数据基础，具有一定对瑜伽动作评估模型探究的推动价值。

(2) 基于神经网络的理论基础，结合当前现有的卷积神经网络、MLP 网络和常见的多模态数据融合方法，实现了多模态数据瑜伽动作识别网络 YogaRegNet 及 5 种体式的多模态数据瑜伽动作评估网络 YogaEvaNet。瑜伽动作识别网络 YogaRegNet 的识别准确率达 99%，在现有的瑜伽动作识别网络中展现出了较高的精度、较优的性能。瑜伽动作评估网络 YogaEvaNet 为目前基于深度学习的瑜伽动作分析研究的一众瑜伽动作识别研究作出了瑜伽动作评估相关的研究补充。同时，这两种瑜伽动作分析网络对多模态数据融合在瑜伽动作分析方面作出了迁移尝试，相较于单一模态数据模型表现出了更好的精度。

5.2 工作展望

本文阐述了瑜伽动作的识别与评估的研究现状及问题，整理标注了瑜伽动作分类评价数据集 YogaCE，研究设计了瑜伽动作识别及评估多模态网络，并进行实验验证。目前相关领域依然面临一定问题，需要广大研究者的不断探索与辛勤工作。在本文实验过程中也发现了一些问题，还存在几点今后可以改进的方向：

(1) 瑜伽动作分类评价数据集 YogaCE 仅涵盖了 5 种常见瑜伽体式，数据量仅 1073 张图片，标

注类别及图片数量都稍显不足，不具有很好的应用广泛性和好的训练效果适用性。今后可以增加数据集的瑜伽体式类别数据量，尽可能涵盖大多数常见的瑜伽动作，使其可以更好地应用到更多瑜伽动作分析研究中。可以通过网络爬虫采集、线下拍摄等方法增加数据集中的图片数量，使通过这一数据集训练的模型可以获得更好的精度和效果。

(2) 瑜伽动作评估网络 YogaEvaNet 在训练过程中测试集损失函数值下降不明显，且最终模型精度不够高，可能出现了网络权重参数未能找到全局最优解的问题。可以在以后研究更多的参数优化策略，优化网络训练。也可能测试训练集的数据量划分并非最优，可以在以后研究其他测试训练集的数据量划分比例，优化网络训练。

(3) 在仅输入骨架坐标数据的 MLP 网络瑜伽动作识别模型中，与双模态同步数据训练获得的仅输入图像数据的 VGG 网络模型相比，大多都精度较低。其中的原因可能是 MLP 这一网络在瑜伽动作数据集上的使用较显落后，或不适用于该数据集骨架坐标数据的训练。针对这一问题，可以考虑在以后的研究中尝试其他更适用的网络，以进一步提升精度。

致 谢

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